ChemicalQDevice | Independent Research Paper | Draft 1.0

La Jolla Move-In: Pancreatic Oncology Clinical Trial Site Complete Documentation Package

15 Documents for California's First PDAC LLM Oncology Clinical Trial Site

Legislation, Regulations, Building Code, Premises Code, Conventional Trial Requirements, and Operations

Draft 1.0 | Repository v4.7.0 | 15 documents | 11 roles | $700,000 per year

01 SB 1188 Site Authorization	02 AB 3162 Patient Rights	03 SB 964 Data Protection	04 H. R. 10412 Federal Act	05 San Diego Municipal Code
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Kevin Kawchak, CEO ChemicalQDevice | kevink@chemicalqdevice.com
10.5281/zenodo.xxxxxxxx | 0009-0007-5457-8667
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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: *This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.*

The paper (v1.0) is at https://doi.org/10.5281/zenodo.xxxxxxxx and the repository (v4.7.0) is at https://github.com/kevinkawchak/physical-ai-oncology-trials. The paper files are deposited in /funding/move-in.

Keywords La Jolla; Pancreatic ductal adenocarcinoma; Phase 1 clinical trial site; Large language model governance; Robotic pancreaticoduodenectomy; Daraxonrasib; Site authorization; Building and premises code; Federal award stewardship; Legislative engagement

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LA JOLLA MOVE-IN, FRONT MATTER

Front Matter: The La Jolla Move-In

1. Why La Jolla, and Why Now

This package stands up one clinical trial site, in La Jolla, San Diego, for one disease and one protocol. It is the successor to the eleven-document San Francisco package, which described a general oncology site operating on demand for twenty-four hours a day and serving 168 unique participants across fifteen cancer types in a single day with twenty-nine robot instances [1, 2].

The La Jolla site is smaller in every dimension that can be counted, and that narrowing is the point rather than a concession. A Phase 1 trial in resectable pancreatic ductal adenocarcinoma treats up to eighteen participants across a 3+3 dose escalation [3]. A facility sized for 168 participants a day would spend its federal award on capacity that the protocol forbids it to use. Every requirement in the thirteen documents that follow the two legislative parts is therefore derived from the protocol, not from a service-line planning ratio.

The choice of La Jolla follows from the partner. The intended partner of choice at the feasibility stage is UC San Diego Moores Cancer Center, which sits in the same community plan area, roughly 1.6 miles from the leasehold described here. The twelve-step feasibility sequence, the sponsor-investigator determination, and the required positioning language are set out in the repository and are carried into this package without softening. No agreement of any kind exists. UC San Diego is not described here as a partner, a sponsor, a trial site, or an endorser, and nothing in this package has been shown to it for approval.

The driver for the move-in is the body of work behind it. Between June 2025 and August 2026 the author deposited twenty papers that together carry a pancreatic oncology trial from drug identification through simulation, regulatory adaptation, protocol, investigational new drug application, and funding [4]. Federal policy moved in the same window: the White House report *Science: A New Golden Age* asks the roughly $200 billion annual federal research portfolio to prioritize the individual scientist over legacy institutions, and its annexed fiscal year 2028 budget priorities memorandum asks agencies for long-duration grants and non-federal cost share [5, 6]. This package is what an independent scientist does with a five-year award once the policy and the evidence are both in hand.

Dimension	San Francisco, 2026	La Jolla, this package
Scope	All oncology, fifteen cancer types	Pancreatic ductal adenocarcinoma only
Operating model	Twenty-four hours, on demand	Extended day, scheduled visits
Participants	168 unique in twenty-four hours	Up to 18 treated, up to 30 screened, over five years
Robots	10 types, 29 instances	8 types, 14 instances
Staff	8 to 10 full-time equivalents	11 roles at 3.95 award-funded full-time equivalents
Documents	11	15
Funding	Not stated	$700,000 per year for five years, $3,500,000 direct
Governing evidence	Simulation	Simulation, a filed protocol, and an investigational new drug application

Table 1. *The predecessor and this package, dimension by dimension. Four rows shrink and three grow, and the growth is in documents, funding, and evidence rather than in capacity.*

2. How This Package Is Organized

Fifteen documents in four parts. Each opens on its own page and can be read alone. Four documents exist here that had no counterpart in the predecessor, because the master prompt asks for three things the San Francisco package never had to answer: conventional pancreatic cancer trial requirements, a lobbying and federal funding position, and a named staff of eleven.

No	Part	Document	What only this document gives the reader
01	I	SB 1188, Site Authorization Act	The state authority under which a site of this kind may exist at all, and the clock the department runs on
02	I	AB 3162, Patient Rights and Robotic Safety Act	The seven participant rights, including the right to a pathway with no model in it
03	I	SB 964, Model Transparency and Data Protection Act	What must be disclosed about the model, and how long a prompt is kept
04	I	H. R. 10412, Federal Act	The federal demonstration designation and the review-timeline reporting requirement
05	II	San Diego Municipal Code Update	Whether this use is permitted on this parcel, and what permit carries it
06	II	Title 22, Division 5, Chapter 15	The department's application, survey, and enforcement machinery
07	II	FDA LLM and Robotic Workflow Compliance Guide	The requirement-by-requirement map from federal law to a document in this package
08	III	Building Code Standards	What has to be built, from floor loading to the machine room
09	III	Premises Code	Who and what may be where, and the robot envelope numbers
10	III	Parking and Transportation Standards	How a participant arrives, and how one leaves after a Whipple
11	III	Emergency Preparedness Plan	What happens when power, network, model, or building fails
12	IV	Site Activation and Standard Operating Procedures	The five gates and the checklist that opens the site
13	IV	Conventional Trial Requirements Manual	Everything the site needs if the model and the robots were deleted
14	IV	Staffing, Role Delineation, and Move-In Plan	The eleven people, what each is for, and the week each arrives
15	IV	Funding Stewardship and Legislative Engagement Plan	Where the money goes, what may be lobbied for, and with whose funds

Table 2. *The fifteen documents. Document 13 is the load-bearing addition: a reader who deletes every model and every robot from the site still holds a complete conventional Phase 1 requirements manual.*

3. The Site at a Glance

The parameters below are fixed once, here. Documents 05 through 14 cite this table rather than restating a dimension, so a single number cannot appear in two documents with two values.

Parameter	Value	Where it is used
Jurisdiction	City of San Diego, La Jolla Community Plan area	Documents 05, 10
Distance to the intended partner	About 1.6 miles	Documents 11, 13
Gross floor area	12,400 square feet	Documents 05, 08
Clinical floor area	7,850 square feet	Documents 08, 09
Robotic procedure suites	2, at 720 square feet each	Documents 08, 09, 13
Procedure suite clear height	10 feet 6 inches	Document 08
Exam rooms	4	Documents 09, 13
Infusion positions	4	Documents 09, 13
Investigational drug pharmacy	340 square feet	Documents 08, 13
Specimen processing laboratory	260 square feet	Documents 08, 13
Machine room	180 square feet	Documents 08, 09, 11
Robot types and instances	8 types, 14 instances	Documents 06, 09, 11
Parking stalls	46	Document 10
Hours of operation	06:00 to 20:00 weekdays; 08:00 to 14:00 Saturday	Documents 05, 12, 14
Participants	Up to 18 treated, up to 30 screened	Documents 10, 13
Trial design	Open-label, single-arm, 3+3 escalation	Document 13
Staff	11 roles, 3.95 award-funded full-time equivalents	Documents 06, 14
Direct award	$700,000 per year, $3,500,000 over five years	Documents 04, 14, 15
Activation target	Week 46 from lease execution	Documents 12, 14

Table 3. *Site parameters, fixed once and cited thereafter. The leasehold address is withheld until a lease is executed; the site is identified by community plan area and by zone in document 05.*

4. The Author Record and the Funding Position

ChemicalQDevice has been a California limited liability company since October 2021. Its first years produced no application: a provisional patent filing that did not convert, forty weeks of seminars, a six-month literature review, and two years of notebook work whose conclusion was that the quantum-inspired approach it had been built around could not escape exponential resource growth as the algorithms scaled. That record is stated here because it is the reason the current method is what it is. The company reached a working method in 2024, when multimodal large language models became able to carry text, image, and document context together, and the first full manuscript that followed tested exactly that capability [7].

Of the fifty-six works in the author's deposited corpus, thirty-six were generated with artificial intelligence assistance in 2026 alone. Twenty of them form the chronology that this trial rests on, from drug identification through protocol, investigational new drug application, and funding [4]. The development practice is public: demonstrations and sessions are run open-source and online alongside other developers, so the method is inspectable rather than asserted.

The funding position assumed throughout this package is a federal award of $700,000 per year for five years, $3,500,000 in direct cost, from at least one federal agency. Favorable responses to project applications have been received from federal funders, three presidential recognition letters have been issued to the chief executive, and industry recipients of the author's communications have been responsive. Each of those facts is stated once more in document 15, with what it does and does not establish. The budget itself is not re-derived anywhere in this package: it is reused from the ten-application work [8].

5. How to Read This Package

Reader	Documents	Why, and in what order
Legislative staff	01, 02, 03, 04	Read 01 first for the definitions the other three inherit, then whichever instrument is in front of you
City planner	05, then 08, 09, 10	05 states the entitlement; the three that follow are the conditions a permit would attach
State surveyor	06, then 12, 14	06 is the survey instrument; 12 and 14 hold the evidence a survey asks for
FDA reviewer	07, then 13	07 maps the site to federal requirements; 13 shows the conventional trial those requirements govern
Funder	Front matter, 13 §7, 14, 15	The evidence, the roster, and the money, in that order
Site principal investigator	13, then 12, 09	The trial first, then the procedures, then the premises rules that constrain them
Prospective participant	02, then 10	The rights that attach to a participant, and how one gets to and from the site

Table 4. *Seven readers and the shortest path through the package for each. No reader is expected to read all fifteen documents, which is why each one repeats the definitions it depends on rather than referring outward.*

DOCUMENT 1 OF 15

SB 1188 - California PDAC LLM Oncology Clinical Trial Site Authorization and Site Establishment Act of 2026

1. Legislative Findings and Declarations

The Legislature finds and declares all of the following.

(a) Pancreatic ductal adenocarcinoma remains among the most lethal solid tumors, and the resectable setting is served by fewer interventional trials than the metastatic setting. The one large readout available in 2026, RASolute 302, reported a median overall survival of 13.2 months against 6.6 in previously treated metastatic disease and is silent on the resectable setting [9].

(b) Simulation work completed between 2025 and 2026 supports, without establishing, a hypothesis that a targeted agent combined with a staged robotic resection may be studied safely in the resectable setting. A ten-arm quantitative systems pharmacology run returned a median overall survival of 12.8 months against 5.4 for chemotherapy, and a 1000-patient digital twin returned a best progression-free survival hazard ratio of 0.31 [10, 11].

(c) That simulation work carries its own stated limitations, which the Legislature adopts rather than sets aside: the pharmacology run assumes no acquired resistance and ideal pharmacodynamics, and the digital twin uses no patient-specific pharmacokinetic, pharmacodynamic, or tumor growth parameters [10, 11].

(d) A credibility assessment aligned to ASME V&V 40 and to ICH M15 scored 81.9 across 55 verification notebook tests, which is a measured figure and not a regulatory finding [11, 12, 13].

(e) Three adapted regulatory frameworks covering good clinical practice, human subject protection, and investigational new drug applications have been published for physical artificial intelligence oncology trials and provide a workable governance spine for a site of this kind [14, 15, 16].

(f) A twenty-four-hour physical artificial intelligence oncology trial simulation demonstrated that a robot fleet can sustain a clinical schedule at 99.7 percent uptime with all seven documented adverse events at Grade 1 or Grade 2 and all resolved within the hour they occurred [2].

(g) California has already legislated in adjacent territory. AB 3030 governs the use of generative artificial intelligence in patient communications, AB 489 addresses deceptive professional terms, SB 1120 governs utilization review, and AB 2013 requires training data transparency [17, 18, 19, 20]. None of the four reaches the interventional clinical trial setting, and none addresses a model whose output is retained as a source record.

(h) A large language model confined to advisory output does not practice medicine and does not replace a clinician. The distinction between advisory output and directive output is the distinction on which participant safety in this setting depends, and it is capable of being written as a testable rule [21].

(i) An independent investigator, operating outside a university, can now assemble a complete regulatory package for a Phase 1 trial, and has done so [3, 22]. State law should not be the reason such a site cannot open.

(j) California has an opportunity to authorize the first such site in the nation. The authorization created by this chapter is conditional, revocable, time-limited, and reportable, because a first authorization that is none of those things is not an authorization but a concession.

2. Definitions

For purposes of this chapter, the following definitions apply. A term defined here carries the same meaning in documents 02, 03, 06, and 07 of this package.

(a) "LLM-directed clinical workflow" means a sequence of clinical tasks in an interventional trial in which a large language model produces output that is presented to a licensed clinician before any action is taken on it.

(b) "Advisory-only output" means output that cannot reach a participant, cannot alter a device setting, cannot change a dose, and cannot enter the clinical record except through an affirmative act by a named human, recorded at the time it is taken.

(c) "Robotic procedure suite" means a room in which one or more robot instances operate within a defined envelope during a study procedure.

(d) "Robot instance" means one physical robot, identified by serial number, registered under §4 of the regulations adopted pursuant to this chapter.

(e) "Site safety officer" means the individual responsible for robot envelopes, interlocks, and the emergency stop chain, who holds the training required by the standard operating procedures of the site.

(f) "Qualified site" means a facility that has met the building, premises, parking, staffing, equipment, and regulatory requirements established under this chapter and holds a current authorization from the department.

(g) "Sponsor-investigator" has the meaning given in 21 CFR 312.3, and the department shall not authorize a site at which a person who controls the sponsor also serves as a clinical investigator [23, 24].

(h) "Audit trail" means a hash-chained, append-only record of every model prompt, every model completion, and every human disposition of a completion, retained under §5 of document 03 of this package.

(i) "Department" means the California Department of Public Health.

3. Site Authorization

(a) An applicant shall file an application containing the contents specified in the regulations adopted under this chapter. The department shall review the application for completeness within 30 days of filing. If the department does not act within 30 days, the application is deemed complete.

(b) The department shall complete substantive review within 90 days of the date an application is complete. If the department does not act within 90 days, a conditional authorization is deemed granted for a period of 180 days, during which the department may complete its review.

(c) A conditional authorization is valid for 24 months and may be renewed for successive 24-month terms.

(d) No participant may be enrolled at a conditionally authorized site until all of the following have occurred: institutional review board approval of the protocol; a safe-to-proceed determination on the investigational new drug application under 21 CFR 312.42 [23]; and a departmental site visit.

(e) Full authorization may be granted after the site has treated six participants and has passed a departmental survey conducted under §5 of the regulations adopted under this chapter.

Step	Action	Statutory clock	Consequence if the department does not act
1	Application filed	Day 0	None; the clock starts
2	Completeness review	30 days	Application deemed complete
3	Substantive review	90 days	Conditional authorization deemed granted for 180 days
4	Conditional authorization issued	24 months	Expires; no enrollment may continue
5	First participant hold released	On three conditions	Hold remains; no enrollment
6	Departmental survey	After 6 participants	Conditional authorization continues to its term
7	Full authorization	24-month renewals	Site reverts to conditional status

Table 5. *The authorization sequence, with the consequence of departmental silence stated for every step. A statute that is silent on inaction has no schedule, only an aspiration.*

4. Operating Conditions

Each condition below is implemented by a document in this package, and the department may verify compliance against that document.

Condition	Implemented by	Evidence the department may request
Minimum staffing and delegation	Document 14	The roster, the delegation of authority log, the training records
Premises zones and robot envelopes	Document 09	The zone plan, the envelope survey, the interlock test record
Model governance and disclosure	Document 03	The model version register and the audit trail extract
Safety reporting	Document 13	The safety report log and the transmission receipts
Emergency response	Document 11	The plan, the quarterly drill records, the after-action reports
Record retention	Document 03	The retention schedule and the archive index

Table 6. *The six operating conditions and the artifact that discharges each. No condition in this chapter is satisfied by a policy statement; each is satisfied by a record.*

5. Inspection, Enforcement, and Suspension

(a) The department may enter and inspect a qualified site during hours of operation without prior notice.

(b) Deficiencies are classified as follows, and the same three classes are used in documents 03, 06, and 12 of this package so that one event is not graded on three scales.

   (1) Class A: a deficiency that presents immediate jeopardy to a participant. The department shall suspend the affected activity immediately, and the civil penalty is not less than $10,000 and not more than $25,000 per occurrence.

   (2) Class B: a deficiency presenting a substantial probability of harm. The site shall file a correction plan within 10 days, and the civil penalty is not less than $2,500 and not more than $10,000.

(3) Class C: an administrative or record deficiency with no probability of harm. The site shall file a correction plan within 30 days, and the civil penalty is not more than $2,500.

(c) A site may appeal a classification to the department within 15 days, and the department shall decide the appeal within 45 days. A Class A suspension is not stayed by an appeal.

6. Reporting and Sunset

(a) The department shall report annually to the Legislature. The report shall state the number of authorized sites, the number of participants enrolled, the number and class of deficiencies cited, every model fault reported under §5 of document 11 of this package, and the department's recommendation on continuation.

(b) This chapter shall remain in effect only until January 1, 2032, and as of that date is repealed, unless a later enacted statute deletes or extends that date. The annual report required by subdivision (a) is the record on which any extension is to be decided.

DOCUMENT 2 OF 15

AB 3162 - California LLM Oncology Patient Rights and Robotic Safety Act of 2026

1. Findings and Declarations

The Legislature finds and declares all of the following. The findings in document 01 §1 of this package are incorporated and are not repeated.

(a) A participant in an interventional oncology trial consents to an investigational agent and to an investigational procedure. Where a large language model also contributes to the conduct of that trial, the participant is entitled to know it, in the consent discussion and not only in the consent form [15].

(b) Patient-facing work published in 2026 establishes that participants understand and use a stated right of refusal when it is offered in plain terms, and that the offer costs a site scheduling capacity rather than safety [25, 26].

(c) A ten-gate assurance suite developed for a mobile surgical humanoid demonstrates that a robot in a clinical envelope can be held to a testable safety specification rather than to a manufacturer's assurance [27].

(d) Clinician trust in a model-assisted workflow is earned by supervision that is recorded, not by model accuracy alone, and the record of supervision is what a participant is entitled to rely on [21].

(e) Existing California law addresses generative artificial intelligence in patient communications and in utilization review, but not the interventional trial setting [17, 19].

(f) An emergency stop that a participant's care team cannot reach within one step is not an emergency stop. Safety in a robotic procedure suite is a matter of measured distances and measured times, and this act states both.

(g) A right that has no evidentiary record cannot be enforced. Each right created by this act is paired in §4 with the record that proves it was honored.

(h) Remedies under this act are limited to injunctive relief, because a damages remedy in an investigational setting would deter the very sites the state is trying to authorize while adding nothing that the department's enforcement power under document 01 §5 does not already provide.

2. Definitions

Participant, advisory-only output, robotic procedure suite, robot instance, and site safety officer carry the meanings given in document 01 §2. In addition:

(a) "Human-only pathway" means the conduct of a participant's trial procedures with no large language model output generated, retrieved, or displayed in connection with that participant's care.

(b) "Robotic operating envelope" means the three-dimensional volume within which a robot instance may move during a study procedure, defined in document 09 §3 and marked on the floor of the suite.

(c) "Emergency stop" means a hardware-initiated command that arrests all robot motion, independent of software state.

(d) "Reportable robotic event" means any unintended motion, any envelope breach, any interlock failure, and any emergency stop actuation, whether or not a participant was present.

3. Participant Rights

A participant at a qualified site has all of the following rights, which may not be waived by contract, by consent form, or by protocol.

(a) The right to a human-only pathway, on request, at any time, without giving a reason and without effect on eligibility, on the assigned dose level, or on any other aspect of trial participation.

(b) The right to be told, in the consent discussion, that a large language model contributes advisory output to the conduct of the trial, what it is used for, and what it is not permitted to do.

(c) The right to know that no model output reaches the participant except through an affirmative act by a named clinician.

(d) The right to a copy of the model output generated in connection with the participant's own care, and of the clinician's disposition of it, on request and at no charge.

(e) The right to be informed of any reportable robotic event that occurred during a procedure in which the participant was present, within 72 hours.

(f) The right to withdraw from the trial without prejudice to continuing care, and to a written statement of the care that will continue.

(g) The right to raise a concern to the department directly, without going through the site, and to be told how.

Subdivision (a) is stated first and without qualification because it is the right that costs the site money. Its operational consequence is carried into the standard operating procedure index at document 12 §3 as a named procedure with a named owner.

4. Disclosure and Consent

Disclosure	Where it is made	Evidence that it was made
A model contributes advisory output	Consent form and consent discussion	Signed consent form, and the coordinator's consent discussion note
What the model is not permitted to do	Consent discussion	The consent discussion note, which names the four prohibitions
The human-only pathway is available	Consent discussion and the participant handout	The handout acknowledgment line, initialed
The audit trail is retained	Consent form	The consent form section on records
A reportable robotic event occurred	Within 72 hours, in writing	The event notification letter and the delivery record
Withdrawal does not end clinical care	Consent form and at withdrawal	The withdrawal note and the continuing care statement

Table 7. *Disclosure, its place, and its evidence. Every row names a record a monitor could ask for, because a disclosure obligation with no record is unverifiable at a survey.*

The consent architecture follows the adapted human subject protection framework rather than being invented here [15, 28]. Where the adapted framework and 45 CFR Part 46 differ, the more protective governs [29].

5. Robotic Safety Requirements

The numbers in this section are identical to document 09 §3, which is the section that owns them. A site may not comply with one and not the other.

Requirement	Value	Standard
Envelope boundary beyond maximum reach	1.2 meters, floor-marked	Site specification
Emergency stop reach from any point inside the envelope	0.8 meters	Site specification
Emergency stop at each suite door	Required	Site specification
Independent interlock channels	3	ISO 13849-1 category 3 [30]
Motion arrest after a stop command	250 milliseconds	IEC 80601-2-77 [30]
Quasi-static force limit in collaborative operation	80 newtons	ISO 13482 [31]
Restart after an emergency stop	Two-person authorization	Document 09 §3

Table 8. *The seven safety values. Each is a measurement a surveyor can take with a tape, a stopwatch, or a force gauge, which is the test this act applies to every safety claim.*

6. Adverse Event Reporting and Remedies

(a) A reportable robotic event shall be recorded in the site's event log within one hour and reported to the department within 5 business days. An event resulting in participant injury shall be reported within 24 hours.

(b) Adverse event reporting to the federal government is governed by the investigational new drug safety reporting requirements and is not displaced by this section [23, 16]. Where the state and federal clocks differ, the shorter governs.

(c) A participant, or the department on a participant's behalf, may bring an action for injunctive relief to enforce a right created by §3. This act creates no right to damages. The limit is stated in the same subdivision as the remedy so that it cannot be read as an oversight.

DOCUMENT 3 OF 15

SB 964 - California Oncology Model Transparency and Clinical Data Protection Act of 2026

1. Findings and Declarations

(a) California has legislated three times on artificial intelligence in health care and once on training data transparency, and none of the four instruments reaches an interventional clinical trial [17, 18, 19, 20].

(b) AB 2013 requires a developer to publish a documentation summary of the data used to train a generative system [20]. It does not require a clinical trial site to say which model version produced a given completion, and in a trial that distinction decides whether a result can be reconstructed.

(c) A prompt and its completion, once a clinician acts on them, are source records. Federal law already requires source records to be attributable, legible, contemporaneous, original, and accurate, and to be retained [32, 23]. This act states how that obligation applies to a model.

(d) A national platform specification published in 2026 established a hash-chained audit trail with provenance services as a workable construction for exactly this record, across sites rather than within one [33].

(e) De-identification obligations under federal health privacy law and the research use limits under the Common Rule both apply, and a site that meets one does not thereby meet the other [34, 29].

(f) A model version that cannot be named cannot be audited. This act therefore makes the version register, not the model, the object of regulation.

2. Definitions

Terms defined in document 01 §2 carry the same meaning here. In addition:

(a) “Model” means a large language model or an ensemble of such models operated on premises at a qualified site.

(b) “Model version” means a specific set of weights, system prompt, retrieval corpus, decoding parameters, and guardrail configuration, identified by a version string and a content hash.

(c) “Prompt” means the complete input presented to a model version, including retrieved context.

(d) “Completion” means the complete output returned by a model version.

(e) “Hash-chained audit trail” means an append-only record in which each entry carries the cryptographic hash of the preceding entry, so that deletion or alteration of any entry is detectable.

(f) “Provenance record” means the linkage from a completion to the model version, the prompt, the retrieval sources, the requesting user, and the human disposition.

(g) “Secondary use” means any use of trial data other than the conduct, analysis, reporting, or regulatory submission of the trial for which it was collected.

3. Model Transparency Obligations

Obligation	Frequency	Record that discharges it
Training data provenance summary	On registration, and on any change	The developer's published documentation summary, filed with the department
Model version register entry	Before first clinical use of a version	The register row: version string, content hash, effective date, approver
Retrieval corpus manifest	Quarterly	The manifest with a content hash per source document
Guardrail configuration statement	On each version	The configuration file hash and a plain-language statement of what is blocked
Known limitation statement	On each version, and on discovery	The limitation log, which carries the date of discovery and the mitigation

Table 9. *Five transparency obligations, each discharged by a record rather than by a policy. The version register is the same artifact the federal change control plan requires under document 07 §5, not a second one.*

A qualified site shall not place a model version into clinical use before its register entry exists. The register is the object the department surveys and the object a federal reviewer is offered, and maintaining one artifact for both is a requirement of this section rather than a convenience.

4. Clinical Data Protection

(a) Trial data shall be de-identified under the expert determination or safe harbor method before any use outside the conduct of the trial [34].

(b) Secondary use is prohibited without specific, separately documented consent. A general consent to research does not authorize secondary use of prompts or completions.

(c) A prompt or completion that contains identifiers is protected health information and is handled as such. A site shall not rely on the fact that a completion was generated rather than recorded to treat it as anything else.

(d) Where the federal privacy standard and the Common Rule differ, the more protective governs, and the site shall document which standard it applied and why [34, 29].

5. Audit Trail and Retention

(a) The audit trail shall record, for each model interaction: the model version string and content hash; the prompt; the completion; the retrieval sources; the requesting user; the timestamp to the second; and the human disposition, which is one of accepted, modified, or rejected, with the name of the clinician who made it.

(b) The audit trail shall be append-only and hash-chained. A site shall verify chain integrity monthly and shall retain the verification record.

(c) Records shall be retained for six years from database lock. Federal law requires two years after the later of marketing application approval or discontinuation of shipment with notice to the agency [23], and state facility record law requires less. Where the periods differ, the longest governs, and six years is the period this chapter sets.

(d) Access logs to any premises zone classified as restricted or machine under document 09 §1 shall be retained for the same six years, so that a question about who was present can be answered for as long as a question about what the model produced.

Record class	Retention	Authority, and why this period
Prompts, completions, dispositions	6 years from lock	Longest of 21 CFR 312.62, state facility law, and this chapter
Model version register	6 years from lock	Must outlive every completion it explains
Chain integrity verifications	6 years from lock	Evidence that the trail was not altered
Premises access logs	6 years from lock	Document 09 §2; the presence question must survive as long as the output question
Consent records	6 years from lock	21 CFR 50 as adapted [15]

Table 10. *Retention by record class. The period is uniform at six years by design: a package whose classes expire on different dates produces an archive that cannot answer a question about a single day.*

6. Enforcement and Breach Notification

(a) A site shall notify the department of a breach affecting trial data within 72 hours of discovery, and affected participants within 15 days. Notification under existing California breach law is not displaced by this section.

(b) Deficiencies under this chapter are classified as Class A, Class B, or Class C with the meanings and penalty ranges given in document 01 §5. A failure of chain integrity that cannot be explained is a Class A deficiency.

(c) A site that places a model version into clinical use without a register entry commits a Class B deficiency for the first occurrence and a Class A deficiency for any subsequent occurrence.

DOCUMENT 4 OF 15

H. R. 10412 - Independent Investigator Pancreatic Cancer Trial Site Act of 2026

1. Short Title and Findings

This Act may be cited as the Independent Investigator Pancreatic Cancer Trial Site Act of 2026. It is the successor to H. R. 9510, the author's own bill, which reached version 5.0 in June 2026 after four full prior iterations and is financially focused: it establishes a cost ledger per computational run and the appropriation machinery behind it [35, 36]. This Act does not restate those financial provisions. It adds the two things H. R. 9510 does not carry: a site designation, and a reporting requirement on federal review timelines for large language model and robotic workflows.

Congress finds the following.

(a) The federal research and development portfolio of roughly $200 billion per year has no systematic framework for identifying where those dollars would catalyze the greatest scientific return, and defers instead to the same institutions that already consume the funding [5].

(b) The fiscal year 2028 research and development budget priorities memorandum asks agencies for long-duration grants, faster award mechanisms, and expanded non-federal cost share so that federal dollars draw in private investment rather than standing alone [6].

(c) Executive Order 14363 launched a national mission that names robotics and physical artificial intelligence as instruments of discovery, and Executive Order 14303 requires that federally supported science meet a stated evidentiary standard [37, 38].

(d) An independent investigator, operating outside a university, has assembled and deposited a complete Phase 1 regulatory package for pancreatic ductal adenocarcinoma, including a protocol, a 172-page investigational new drug application, and adapted good clinical practice, human subject protection, and investigational new drug frameworks [3, 22, 14, 15, 16].

(e) No federal mechanism currently designates a clinical trial site as a demonstration site for a large language model and robotic workflow, and no federal report states how long the review of such a workflow takes. What is not measured is not managed.

(f) A demonstration designation that carries no money and no exemption still has value, because it fixes a point of contact, a review clock, and a public record.

2. Definitions

In this Act:

(a) The term "independent investigator" means a natural person, or a small business concern controlled by a natural person, that is not an institution of higher education and not a controlled affiliate of one.

(b) The term "designated demonstration site" means a clinical trial site designated under section 101.

(c) The term "LLM-directed clinical workflow" has the meaning given in document 01 §2 of the package in which this Act is published, and is restated here so that this Act stands alone: a sequence of clinical tasks in an interventional trial in which a large language model produces output that is presented to a licensed clinician before any action is taken on it.

(d) The term "covered review action" means any action by the Secretary on an investigational new drug application, an investigational device exemption, or a pre-submission request that concerns an LLM-directed clinical workflow.

3. Demonstration Site Designation

Sec	Provision	What it obligates, and of whom
101	Designation authority	The Secretary may designate not more than 5 sites per fiscal year, on application
102	Eligibility criteria	An applicant must hold a state authorization, an active protocol, and a safe-to-proceed determination
103	Term and renewal	5 years, renewable once, with a public record of each designation
104	No exemption created	Designation confers no waiver of any requirement of title 21, United States Code, or of its regulations
105	Independent investigator set-aside	Not fewer than 2 of the 5 annual designations reserved for independent investigators
201	Authorization of appropriations	$3,500,000 per designated site over 5 years, not to exceed $17,500,000 per fiscal year cohort
301	Review timeline reporting	The Secretary reports annually on covered review actions and their elapsed times
401	Savings clause	Nothing in this Act displaces 21 CFR parts 50, 312, or 812, or the Federal Policy for the Protection of Human Subjects

Table 11. *The operative sections. Section 104 is the provision that makes the rest of the Act defensible: a demonstration designation that waived a safety requirement would be a worse instrument than none at all.*

4. Authorization of Appropriations

The authorization is sized directly against the frame already established for a single site: $700,000 per year of direct cost and $3,500,000 over five years, with no cost share and no program income anticipated [8, 39]. That figure is reused here and is not re-derived.

Quantity	Per site	Per cohort	Basis
Annual direct cost	$700,000	$3,500,000	Document 15 §1
Five-year direct cost	$3,500,000	$17,500,000	Five times the annual line
Sites per fiscal year cohort	1	5	Section 101
Independent investigator set-aside	n/a	2 of 5	Section 105
Five cohorts, ten years	n/a	$87,500,000	Five cohorts at $17,500,000

Table 12. *The appropriation arithmetic, shown rather than asserted. Five cohorts at five sites each is the whole program, and it is smaller than a single Phase 3 trial.*

5. FDA Review Timelines and Reporting

Section 301 is the operative provision for the purpose this Act was written to serve: making a large language model and robotic workflow reviewable on a published schedule. Not later than one year after enactment, and annually thereafter, the Secretary shall submit to the Congress a report that states, for the preceding fiscal year:

(a) the number of covered review actions received, by type;

(b) the median and ninetieth-percentile elapsed time from receipt to action, by type;

(c) the number of clinical holds imposed on an application involving an LLM-directed clinical workflow, and the stated basis for each;

(d) the number of pre-submission meetings requested and granted, and the median time to meeting; and

(e) any guidance the Secretary intends to issue on the subject in the following fiscal year.

The report requires no new authority and no new inspection. Every quantity in it already exists inside the agency's own tracking, and the adapted investigational new drug framework published in 2026 identifies each of them by regulation citation [16, 23].

6. Relationship to Existing Law

Nothing in this Act may be construed to displace, waive, or modify any requirement of 21 CFR part 50 (protection of human subjects), 21 CFR part 54 (financial disclosure by clinical investigators), 21 CFR part 312 (investigational new drug application), 21 CFR part 812 (investigational device exemptions), or the Federal Policy for the Protection of Human Subjects [28, 24, 23, 40, 29]. Designation under section 101 is a statement that a site has met existing requirements, and never a statement that it need not.

DOCUMENT 5 OF 15

San Diego Municipal Code Update - PDAC LLM Oncology Clinical Trial Site Requirements, La Jolla Community Plan Area

1. Purpose and Applicability

The purpose of this update is to establish the land use status, the permit pathway, and the operating standards for a pancreatic oncology clinical trial site incorporating an LLM-directed clinical workflow and robotic procedure suites, within the La Jolla Community Plan area of the City of San Diego. It applies to a facility that holds or has applied for a state authorization under document 01 of this package and that does not hold a general acute care hospital license.

The site is identified in this package by community plan area and by zone. The street address is withheld until a lease is executed, because publishing a parcel before terms are agreed prices the parcel. Nothing in this document should be read as a representation that any specific parcel has been secured; none has [41].

2. Zoning and Land Use

Zone	Use status	Permit	Condition attached by this update
Commercial office	Permitted	Neighborhood development permit	Robot envelopes wholly interior; no exterior mechanical addition above the parapet
Commercial community	Permitted	Neighborhood development permit	Loading and waste pickup limited to the hours in §5
Mixed use, commercial ground floor	Conditionally permitted	Conditional use permit, process 3	Acoustic separation from any residential use above or adjacent
Residential, any density	Not permitted	None available	Not permitted in any residential zone in the community plan area
Open space and coastal bluff	Not permitted	None available	Not permitted

Table 13. *Zone by zone. Two zones permit the use outright, one permits it conditionally, and two do not permit it at all. A use that is not permitted anywhere is stated as such rather than omitted.*

The use is classified as a medical office and outpatient procedure use with an accessory research function. It is not classified as a hospital, and the site does not provide inpatient beds, an emergency department, or overnight stays. A participant whose condition requires inpatient care is transferred under document 11 §2.

3. Conditional Use Permit Findings

Where a conditional use permit is required, the hearing officer shall make each of the following findings.

(a) The proposed use is consistent with the La Jolla Community Plan and the General Plan.

(b) The proposed use will not adversely affect the applicable land use plan.

(c) The proposed use will not be detrimental to the public health, safety, and welfare.

(d) The proposed use complies with the regulations of the Land Development Code, including any allowable deviations pursuant to the Code.

(e) Every robotic operating envelope, as defined in document 09 §3, lies wholly within the building envelope, and no robot instance operates in a location visible from, or reachable from, the public right of way.

(f) The applicant has filed a model governance record consisting of the model version register required by document 03 §3, and has agreed to make it available to the City on request in connection with any complaint.

Findings (e) and (f) are specific to this use. The first four are the findings the City already makes for any conditional use, and are restated here so that a hearing officer reading this document alone has the complete set.

4. Coastal Overlay and Community Plan Consistency

Much of the La Jolla Community Plan area lies within the Coastal Overlay Zone. The following applies where a proposed site does.

(a) A tenant improvement wholly inside an existing permitted structure, with no increase in building height, no increase in gross floor area, no change to the exterior envelope, and no reduction in public access, is an improvement that does not intensify a coastal resource impact.

(b) The applicant shall nonetheless file for a coastal development permit where the City determines that the change of use, as distinct from the construction, triggers one under the Coastal Act [42]. This document does not assert that no permit is required; it states the analysis the applicant must complete and file.

(c) Review under the California Environmental Quality Act proceeds on the existing facilities exemption where the project involves no expansion of use beyond that existing at the time of the determination, and on a categorical exemption for minor alterations otherwise. The applicant shall state on the record which exemption is relied on and shall file the supporting analysis [43].

(d) Where a proposed site lies outside the Coastal Overlay Zone, subdivisions (a) through (c) do not apply and the ordinary permit sequence in §6 governs.

5. Operating Standards

Standard	Limit	Basis and cross-reference
Hours of operation	06:00 to 20:00 weekdays; 08:00 to 14:00 Saturday; closed Sunday	Front matter site table; documents 12 §1 and 14 §4 use the same window
Sunday and after-hours exception	A procedure already in progress, or a participant safety event	Document 11 §1
Noise at the property line	55 dBA daytime, 50 dBA evening	Mechanical equipment is the only exterior source; the robots are interior
Exterior lighting	Full cutoff, 3000 K maximum, off by 22:00 except at exits	Coastal and residential adjacency
Delivery window	07:00 to 17:00 weekdays	Loading geometry in document 10 §4
Regulated medical waste pickup	Twice weekly, Tuesday and Friday, 07:00 to 09:00	Waste streams in document 09 §5
Exterior queuing	Prohibited; all waiting is interior	Document 10 §4

Table 14. *Seven operating standards. The hours are the single most consequential row: this is an extended-day site, not the twenty-four-hour facility described in the San Francisco predecessor, and the difference follows from a protocol that schedules visits rather than accepting them on demand.*

6. Permits, Fees, and Plan Check

The permit sequence below determines the move-in schedule in document 14 §6, which is why the parallel and sequential relationships are stated here rather than assumed.

Step	Permit or review	Relationship	Depends on
1	Zoning verification	First	Nothing
2	Neighborhood development or conditional use permit	Sequential	Step 1
3	Coastal development permit, if required	Parallel with 4	Step 1
4	Building plan check	Sequential	Step 2
5	Mechanical, electrical, plumbing	Parallel with 4	Step 2
6	Fire plan check	Parallel with 4	Step 2
7	Hazardous materials business plan	Parallel with 4	Step 2
8	County health plan check	Sequential	Steps 4 through 7
9	Certificate of occupancy	Last	Step 8 and commissioning under document 08 §6

Table 15. *Nine permits in sequence. Four of the nine run in parallel, which is the only reason the forty-six-week move-in schedule in document 14 §6 closes; a strictly sequential reading of the same nine adds roughly eleven weeks.*

DOCUMENT 6 OF 15

California Code of Regulations, Title 22, Division 5, Chapter 15 - PDAC LLM Oncology Clinical Trial Site Regulations

1. Scope and Authority

This chapter is adopted under the authority of document 01 of this package and the licensing provisions of the Health and Safety Code [44, 45]. It governs the authorization, survey, and enforcement of a pancreatic oncology clinical trial site incorporating an LLM-directed clinical workflow.

This chapter does not govern a general acute care hospital, a clinic licensed under other chapters of this division, or a research laboratory that enrolls no human participants. Where a facility holds another license, this chapter applies only to the trial activity described in document 01 §2 and does not displace the requirements of that license.

2. Application for Authorization

An application shall contain each of the following. Each item is an artifact that exists elsewhere in this package, so that an applicant is never asked for a document that has not been defined.

Item	Content	Source	Form in which it is filed
(a)	Staffing plan and delegation log	Document 14	Roster, qualifications, signed delegation entries
(b)	Premises plan with zones and envelopes	Document 09	Floor plan with zone boundaries and envelope markings
(c)	Building commissioning record	Document 08 §6	Commissioning report and certificate of occupancy
(d)	Emergency preparedness plan	Document 11	Plan, drill schedule, mutual aid letter
(e)	Model version register	Document 03 §3	Register extract with content hashes
(f)	Robot instance register	§4 of this chapter	Serial numbers, envelope, last interlock test
(g)	Protocol and regulatory status	Document 13 §2	Protocol, institutional review board approval, safe-to-proceed record
(h)	Standard operating procedure index	Document 12 §3	Index with owners and review dates
(i)	Parking and transportation plan	Document 10	Stall schedule and the transportation demand plan

Table 16. Nine application items. A regulation whose application asks for documents that do not exist cannot be complied with, so every row names the document in this package that produces the artifact.

3. Staffing and Qualification Requirements

Position	Minimum	Qualification the department will verify
Site principal investigator	0.10 FTE	California medical license; hepatopancreatobiliary surgical practice; 40 hours investigator training
Sub-investigator, medical oncology	0.10 FTE	California medical license; gastrointestinal medical oncology practice; 40 hours investigator training
Director of clinical operations	0.40 FTE	Five years of trial operations; 80 hours coordinator training
Lead clinical research coordinator	1.00 FTE	Coordinator certification; 80 hours training; consent competency assessment
Regulatory affairs and quality manager	0.45 FTE	Three years of regulatory submissions; 80 hours training
Investigational drug pharmacist	0.20 FTE	California pharmacist license; sterile compounding competency under USP 797 and 800
Site safety officer	0.55 FTE	160 hours safety training; robot platform certification; interlock test competency
Model governance lead	0.40 FTE	120 hours training; verification and validation competency; audit trail administration
Data manager and biostatistician	0.30 FTE	Statistical qualification; database lock competency
Research nurse and participant navigator	0.25 FTE	California registered nurse license; oncology experience; 80 hours training

Table 17. *Staffing minimums by position. The full-time equivalent column is identical to the roster in document 14 §1 and the training hours are identical to document 12 §5; the department surveys against those two documents rather than against a separate standard.*

The chief executive and sponsor representative is not listed above, because that role is not a clinical position and the department does not verify a qualification for it. Its exclusion is deliberate and is explained in document 14 §3.

4. Robot and Model Registration

(a) A site shall maintain a robot instance register with one row per physical robot: manufacturer, model, serial number, suite assignment, envelope identifier, date of last interlock test, and date of last envelope survey. The site described in this package registers 14 instances across 8 types.

(b) A site shall maintain a model version register under document 03 §3, with one row per model version placed into clinical use.

(c) An amendment to either register shall be filed with the department within 10 business days of the event that triggers it. The triggering events are: addition or removal of a robot instance; relocation of an instance between suites; a change to an envelope; a failed interlock test; and the placement of a new model version into clinical use.

(d) A robot instance that has failed an interlock test shall not be used in a study procedure until it has passed a retest and the register has been amended.

5. Survey, Deficiency Classes, and Penalties

(a) The department shall survey each authorized site at least annually, and may survey without notice at any time during the hours of operation stated in document 05 §5.

(b) Deficiencies are classified as Class A, Class B, or Class C with the meanings, correction periods, and civil penalty ranges established in document 01 §5. Those definitions are not restated here and are not varied.

(c) The department shall issue a statement of deficiencies within 15 business days of the survey exit conference. The site shall file its correction plan within the period the class requires.

(d) An appeal is taken under document 01 §5(c). A Class A suspension is not stayed by an appeal.

6. Records and Reporting

(a) A site shall file an annual report stating: participants screened, enrolled, and treated; every reportable robotic event under document 02 §2; every model fault under document 11 §5; every deficiency cited and its correction; and the current registers.

(b) Records shall be retained for six years from database lock, which is the period fixed by document 03 §5. Where this chapter, federal law, and the facility record law of this state prescribe different periods, the longest governs.

(c) A site shall notify the department within 24 hours of any event that would constitute a Class A deficiency, whether or not the department has cited it. Self-reporting within 24 hours is a mitigating factor in the civil penalty determination and shall be stated as such in the department's order.

DOCUMENT 7 OF 15

FDA LLM and Robotic Workflow National Compliance Guide for a Phase 1 PDAC Clinical Trial Site

1. Purpose and Scope

This document is a map from one clinical trial site to the federal requirements that already govern it. It is not a submission, not a guidance document, not an agency position, and not a claim that any workflow described here has been reviewed or accepted. The honest form of the request that a large language model and robotic workflow be acceptable to the agency is a map that a reviewer can check, requirement by requirement, against artifacts that exist.

Its scope is the site described in the front matter of this package, operating the Phase 1 protocol at document 13 §2. Nothing here applies to a marketed device, to a commercial software product, or to any use of a model outside an investigational setting.

2. The Regulatory Position of the Site

The trial is a combined investigational new drug and investigational device investigation [3]. The filing is the 172-page investigational new drug application deposited in July 2026, which follows the section skeleton established by the adapted Part 312 framework [22, 16].

Component	Pathway	Why it sits there
Perioperative daraxonrasib	Investigational new drug, 21 CFR 312	An investigational agent administered under a protocol [23]
Robotic surgical platform	Investigational device exemption, 21 CFR 812, or a cleared device used within its labeling	Determined at the site agreement; the configuration is not specified before then
LLM advisory output	Neither, while it remains advisory-only	It produces no output that reaches a participant except through an affirmative human act
Electronic records and signatures	21 CFR 11	Prompts, completions and dispositions are source records [32]
Financial disclosure	21 CFR 54	Decides the sponsor-investigator firewall in document 14 §3 [24]
Human subject protection	21 CFR 50 and 45 CFR 46	Consent and protection, as adapted [28, 29, 15]

Table 18. *Six components and the pathway each sits on. The third row is the one that must be defended: the advisory-only boundary in §5 is what keeps the model out of the device pathway, and if the boundary moves, the row moves with it.*

3. Correction Map to 21 CFR Parts 11, 50, 54, 312, and 812

Citation	Requirement	Where this site discharges it
21 CFR 11.10(a)	Validation of systems to ensure accuracy and consistency	Document 12 §6 quality management; the model version register at document 03 §3
21 CFR 11.10(e)	Secure, computer-generated, time-stamped audit trails	Document 03 §5, hash-chained and verified monthly
21 CFR 11.10(d)	Limiting system access to authorized individuals	Document 09 §2 access control; the machine room is a restricted zone
21 CFR 50.20	General requirements for informed consent	Document 02 §4 and document 13 §3
21 CFR 50.25	Elements of informed consent	Document 02 §4, including the model disclosure and the human-only pathway
21 CFR 54.2	Definitions triggering financial disclosure	Document 14 §3, which assigns the investigator role wholly to the site
21 CFR 54.4	Certification and disclosure statements	Document 15 §2 stewardship controls
21 CFR 312.23	Investigational new drug content and format	The filed application [22]
21 CFR 312.32	Safety reporting	Document 13 §6, with the state clock in document 02 §6
21 CFR 312.42	Clinical holds	Document 01 §3(d), the first participant hold
21 CFR 312.57	Sponsor recordkeeping for drug disposition	Document 13 §5 pharmacy
21 CFR 312.62	Investigator recordkeeping and retention	Document 03 §5, at six years
21 CFR 812.140	Device records	Document 06 §4 robot instance register
21 CFR 812.150	Device reports	Document 02 §6 reportable robotic events

Table 19. *Fourteen requirements and the document in this package that discharges each. Every row names a document rather than a policy, so a reviewer can ask for the artifact by name.*

4. ICH E6(R3) Mapping

The good clinical practice spine is the adapted ICH E6(R3) framework published in March 2026 rather than the source guideline, because the adaptation addresses the physical artificial intelligence case the source does not [14, 46]. Where the two differ, the difference is stated.

Principle	Site implementation	Where the adaptation differs from the source
Quality by design	Document 12 §6	Adds the model version as a quality-critical factor
Investigator responsibility	Document 14 §2	Names a delegated task set for a model governance role the source does not contemplate
Data governance	Document 03 §5	Treats a completion as a source record
Risk-proportionate monitoring	Document 13 §6	Adds envelope and interlock records to the monitoring scope
Records and reports	Document 03 §5	Extends retention to six years

Table 20. *Five good clinical practice principles. Each row states where the adapted framework goes beyond the source, because a reviewer who is told only that a site follows good clinical practice learns nothing about the model.*

5. The Advisory-Only Boundary and Software Change Control

This is the load-bearing section of the document. The boundary is written as a rule that can be tested rather than as a commitment.

(a) **What the model may produce.** A summary of source documents already in the record; a proposed differential; a proposed order set; a proposed eligibility assessment; a draft narrative for an adverse event; and a literature retrieval with citations.

(b) **What the model may not do.** It may not transmit any output to a participant. It may not write to the electronic data capture system. It may not set, change, or confirm a dose. It may not command, configure, or release any robot motion. These four prohibitions are enforced at the interface, not by policy: the model process has no write credential to the data capture system and no route to the robot control network, which is segmented under document 08 §5.

(c) **What a human must do first.** A licensed clinician named in the delegation log must record a disposition of accepted, modified, or rejected before any completion is acted on. A completion with no recorded disposition is treated as rejected.

(d) **What record proves it.** The audit trail entry under document 03 §5, carrying the model version, the prompt, the completion, the clinician name, the disposition, and the timestamp to the second.

Verification precedes generation: the site does not accept a completion into the record on the strength of the model's confidence, and the verification-first position is the one established in the 2026 legislative and assurance work [36, 21].

Software change control follows a predetermined change control plan [47]. A change to weights, system prompt, retrieval corpus, decoding parameters, or guardrail configuration is a new model version and requires a new register entry under document 03 §3 before clinical use. The register is a single artifact serving both the state obligation and the federal plan, which is stated in document 03 §3 and is repeated here so that neither reader is surprised.

6. Pre-Submission Sequence and Evidence Package

Step	Action	Evidence carried, and the limitation stated with it
1	Pre-submission request [48]	The protocol, the boundary rule in §5, and the segmentation drawing
2	Credibility package	ASME V&V 40 and ICH M15 aligned score of 81.9 across 55 verification tests [11, 12, 13]. Limitation: no patient-specific pharmacokinetic, pharmacodynamic, or tumor growth parameters
3	Simulation evidence	Ten-arm run at 12.8 months median overall survival against 5.4 [10]. Limitation: assumes no acquired resistance and ideal pharmacodynamics
4	External comparison	RASolute 302 at 13.2 against 6.6 [9]. Limitation: metastatic and previously treated; silent on the resectable setting
5	Model documentation	The version register, the guardrail statement, and the change control plan [47, 49]
6	Investigational new drug submission	The filed application [22]

Table 21. *The six-step sequence. Every evidence row carries the limitation its own source stated, in the same cell as the result, because a reviewer who has to look elsewhere for the caveat has been given a claim rather than evidence.*

DOCUMENT 8 OF 15

Building Code Standards for a PDAC LLM Oncology Clinical Trial Facility

1. Scope, Occupancy, and Applicable Codes

This document states what has to be built. It applies to a 12,400 gross square foot leasehold with 7,850 square feet of clinical floor area, within an existing permitted structure in the La Jolla Community Plan area.

Area	Occupancy	Consequence of the classification
Robotic procedure suites, recovery, sterile core	Ambulatory care, Group B with ambulatory care provisions	Sprinklers throughout the smoke compartment; two means of egress from each suite; smoke compartment not exceeding 22,500 square feet
Exam rooms, infusion positions	Business, Group B	Standard egress; no additional smoke compartmentation
Investigational drug pharmacy	Business, Group B with hazardous materials control area	Containment and exhaust under USP 800; quantities below control area maximums [50]
Specimen laboratory	Business, Group B	Negative pressure; no flammable storage above control area limits
Machine room	Business, Group B	Dedicated cooling; restricted access under document 09 §2
Administrative and staff	Business, Group B	Standard

Table 22. *Occupancy by area. The first row is the expensive one: the ambulatory care classification, not the robots, drives the sprinkler, egress, and smoke compartment cost in the tenant improvement budget at document 14 §6.*

The governing codes are the California Building Standards Code, Title 24 [51]; NFPA 99 for health care facilities [52]; and NFPA 101 for life safety [53]. Where the California amendments and the model code differ, the California amendment governs.

2. Site and Structural Provisions

Element	Requirement	Why this trial needs it
Procedure suite live load	100 pounds per square foot	Robot bases, boom-mounted equipment, and a full instrument cart on the same slab
Robot base anchorage	Post-installed anchors to a slab not less than 5 inches, designed for the manufacturer's overturning moment	Registration drift is a data quality failure before it is a safety failure
Vibration at the robot base	Not more than 3,200 micro-inches per second, one-third octave, during operation	Instrument tip deflection is the limiting factor for a needle-placement instance
Procedure suite clear height	10 feet 6 inches to the lowest obstruction	Boom-mounted lights and the surgical platform's full articulation
Suite floor area	720 square feet each, two suites	Envelope plus the 1.2 meter clearance in document 09 §3 plus a circulation path
Seismic restraint	All suspended equipment, medical gas, and the machine room racks	Title 24 seismic provisions; post-event inspection under document 11 §3

Table 23. *Six structural provisions. The vibration limit is the one a general contractor will question, and it is stated because the device envelope work that this site inherits assumes it [54, 27].*

3. Mechanical, Electrical, and Plumbing

Space	Air changes	Pressure	Additional requirement
Robotic procedure suite	20 per hour	Positive, 0.01 inches water column	Minimum 4 outside air changes; temperature 68 to 75 degrees Fahrenheit
Sterile core	20 per hour	Positive	Adjacent to both suites
Soiled utility	10 per hour	Negative	Exhausted directly outdoors
Pharmacy buffer room	30 per hour	Positive	ISO Class 7; USP 797 [50]
Pharmacy containment area	12 per hour	Negative	External exhaust; USP 800 [50]
Specimen laboratory	8 per hour	Negative	Fume hood exhausted directly outdoors
Machine room	6 per hour	Neutral	64 to 81 degrees Fahrenheit; dedicated cooling

Table 24. *Air changes and pressure by space. Document 09 §4 sets traffic patterns against exactly these relationships, and document 13 §5 sets the pharmacy and laboratory requirements against the same two rows.*

The essential electrical system carries three branches under NFPA 99: life safety, critical, and equipment [52]. Both procedure suites, the machine room, the specimen freezers, and the pharmacy hoods are on the critical or equipment branch. Isolated power is provided in each procedure suite. The plumbing serving the pharmacy buffer and containment areas is separated from general plumbing, with a dedicated hand-washing sink outside the buffer room boundary.

4. Medical Gas, Shielding, and Specialty Systems

Each procedure suite is provided with oxygen, medical air, nitrogen for instrument drive where the platform requires it, and two waste anesthetic gas disposal connections. Vacuum is provided at four outlets per suite. Medical gas systems are installed, tested, and verified under NFPA 99, and the verification record is part of the commissioning package in §6 [52].

The protocol schedules cross-sectional imaging for response assessment under RECIST 1.1 at intervals stated in document 13 §5 [55]. That imaging is performed off site under the imaging agreement, so this facility provides no computed tomography or magnetic resonance suite and requires no structural shielding for one. Intraoperative ultrasound is performed with a mobile imaging assistant instance and requires no shielding. Stating this explicitly is deliberate: a reader who assumes an imaging suite will over-build the mechanical and electrical systems by a wide margin.

5. Technology Infrastructure and the Machine Room

The on-premises model node is why this document differs from an ordinary ambulatory surgical center building code. The protocol requires the model to run on premises rather than on a hosted service [3].

Provision	Requirement	Cross-reference
Room area	180 square feet	Front matter site table
Power	Two 208 volt three-phase circuits on the equipment branch	NFPA 99 [52]
Uninterruptible supply	30 minutes at full load, then generator	Document 11 §4
Cooling	Dedicated, redundant, 64 to 81 degrees Fahrenheit	§3 of this document
Physical access	Restricted zone; badge plus second factor	Document 09 §2
Network segmentation	Three segments: clinical, model, robot control, with no route between model and robot control	Document 07 §5, and document 11 §5
Egress of data	One monitored path to the audit trail store; no outbound internet from the model segment	Document 03 §5

Table 25. *Seven machine room provisions. The sixth is the one that makes the advisory-only boundary in document 07 §5 a matter of wiring rather than of policy: there is no route from the model segment to the robot control segment.*

6. Commissioning and Certificate of Occupancy

Commissioning proceeds as a gated sequence. The last commissioning gate is the first activation gate in document 12 §1, so the two documents share a boundary rather than overlapping.

Gate	Scope	Evidence that closes it
C1	Structural and anchorage	Special inspection reports; anchor pull test records
C2	Mechanical air balance	Balance report showing every row of Table 24 met
C3	Electrical and essential branch	Generator load bank test; transfer time record; isolated power line isolation monitor test
C4	Medical gas	NFPA 99 verification certificate [52]
C5	Fire and life safety	Sprinkler acceptance; alarm test; egress inspection [53]
C6	Machine room and network	Segmentation test showing no route from model to robot control; cooling redundancy test
C7	Robot installation and envelope survey	Envelope markings surveyed; interlock test record for all 14 instances
C8	Certificate of occupancy	The certificate, issued after the county health plan check under document 05 §6

Table 26. *Eight commissioning gates. Gate C6 is tested by attempting a connection and failing to make one, which is the only form of evidence that a segmentation claim can carry.*

DOCUMENT 9 OF 15

PDAC LLM Oncology Clinical Trial Site Premises Code

1. Scope and Zone Classification

This document states who and what may be where. The premises are divided into four zones. The zone scheme fixed here is used unchanged by documents 10, 11 and 12.

Zone	Rooms	Who may enter	Credential and record
Public	Lobby, waiting, public restrooms, participant education room	Anyone, during hours of operation	None; closed-circuit recording retained 90 days
Clinical	Exam rooms, infusion positions, phlebotomy, consultation	Participants with an appointment, escorted visitors, clinical staff	Badge for staff; sign-in for participants; visit record
Restricted	Procedure suites, sterile core, soiled utility, pharmacy, specimen laboratory	Named staff on the delegation log; a participant during a scheduled procedure	Badge plus role authorization; access log retained six years
Machine	Machine room, network closets, audit trail store	Model governance lead, site safety officer, and named information technology staff	Badge plus second factor; two-person entry for any change; access log retained six years

Table 27. *The four premises zones. A zone with no access record is a zone with no control, so every row names both the credential and the record, and the two restricted-zone retention periods match document 03 §5.*

2. Access Control and Physical Security

(a) Badges are issued in four classes matching the four zones. A badge grants the lowest zone required for the holder's delegated tasks and no higher.

(b) A visitor to the clinical zone is escorted at all times by a badged staff member. A visitor may not enter the restricted or machine zones.

(c) After-hours access, meaning access outside the hours in document 05 §5, is granted only for a procedure already in progress or a participant safety event, and generates an alert to the director of clinical operations.

(d) Access logs for the restricted and machine zones are retained for six years from database lock, the period fixed by document 03 §5, so that a question about who was present can be answered for as long as a question about what the model produced.

(e) Closed-circuit recording covers the public zone, the loading area, and every restricted zone door, and does not cover the interior of an exam room, a procedure suite, or a restroom.

3. Robot Operating Envelopes and Interlocks

This section owns the numbers. Document 02 §5 restates them and may not vary them.

Requirement	Value	How it is verified
Envelope boundary beyond maximum reach	1.2 meters	Surveyed and floor-marked at commissioning gate C7
Emergency stop reach from any point inside the envelope	0.8 meters	Measured with a tape at each survey
Emergency stop at each suite door	Required	Visual inspection and actuation test
Independent interlock channels	3, at category 3	Channel-by-channel fault injection [30]
Motion arrest after a stop command	250 milliseconds	High-speed timing record [30]
Quasi-static force limit, collaborative operation	80 newtons	Force gauge at the instrument tip [31]
Restart after an emergency stop	Two-person authorization	Log entry with both names, and a re-verification of the register under document 06 §4
Interlock test interval	Monthly, and after any fault	Test record filed to the robot instance register

Table 28. *Eight envelope and interlock requirements. Each is verified by a measurement rather than by a manufacturer's assurance, which is the position the ten-gate assurance work established [27].*

The eight robot types and fourteen instances are distributed as follows: two robotic surgical platforms, one per suite; two limb-stabilization collaborative instances; one imaging-guided needle placement instance; two intraoperative ultrasound imaging assistants; two specimen transport and handling instances; one pharmacy compounding instance; two rehabilitation gait trainers; and two sterile processing loading instances.

4. Clean, Sterile, and Soiled Corridors

Traffic follows the pressure relationships fixed in document 08 §3 and does not vary from them.

(a) Sterile supply moves from the sterile core into a procedure suite and never in the reverse direction.

(b) A used instrument tray moves from a procedure suite into soiled utility through the soiled corridor, and never through the sterile core.

(c) A specimen transport instance moves between a procedure suite and the specimen laboratory on a defined path that does not cross the sterile core. The path is marked and is part of the envelope survey at commissioning gate C7.

(d) Where the geometry of the leasehold forces a single corridor to serve both directions, a temporal separation is used instead: soiled movement is confined to a scheduled window after the last procedure of the day, and the schedule is a controlled record.

5. Waste Streams and Housekeeping

Stream	Container	Pickup	Record
General	Standard, lidded	Twice weekly	None required
Recyclable	Standard, labeled	Weekly	None required
Regulated medical	Red, rigid, lined	Twice weekly, Tuesday and Friday, 07:00 to 09:00	Manifest retained three years
Cytotoxic and pharmaceutical	Yellow, rigid, sealed	Weekly, by the licensed hauler	Manifest retained three years; USP 800 handling [50]
Electronic and media	Locked, tamper-evident	On demand	Destruction certificate; media from the machine zone destroyed rather than resold

Table 29. *Five waste streams. The pickup windows are identical to document 05 §5, because a municipal operating standard and a premises procedure that disagree about a delivery window produce a code violation on an ordinary Tuesday.*

Housekeeping in the restricted zone is performed by trained staff on a written schedule, using agents compatible with the robot manufacturers' cleaning instructions. Terminal cleaning of a procedure suite follows every procedure and is recorded.

6. Signage, Wayfinding, and Occupant Conduct

(a) Every zone boundary carries signage stating the zone, the credential required, and the name of the responsible role.

(b) Each procedure suite door carries the emergency stop location, the envelope boundary notice, and the two-person restart requirement.

(c) The wayfinding path from the accessible parking stall to the clinical zone entrance is the accessible route described in document 10 §3, and is signed at each decision point.

(d) No person may enter a marked envelope while a robot instance in that envelope is energized, except a member of the procedure team performing a delegated task. A participant is positioned before energizing and is never moved into an energized envelope.

(e) Photography and recording are prohibited in the restricted and machine zones without written authorization from the director of clinical operations.

DOCUMENT 10 OF 15

PDAC LLM Oncology Clinical Trial Site Parking and Patient Transportation Standards

1. Scope and Applicability

This document governs parking supply, the accessible route, arrival and departure geometry, and the transportation a participant needs after a pancreaticoduodenectomy. It applies to the leasehold described in the front matter and to the shared structure serving it.

Where the City of San Diego parking minimum for a medical office use and the requirement derived here differ, the higher governs. That rule is stated once, here, and is not repeated in document 05 [41].

2. Stall Count, Mix, and Dimensions

The participant stall count is derived from the visit schedule rather than from a floor area ratio. A Phase 1 site treating up to eighteen participants has a peak concurrency that can be counted, and counting it is more defensible before a hearing officer than applying a general medical office ratio to 12,400 square feet.

Stall class	Count	Dimension	Basis for the count
Participant	22	8.5 by 18 feet	Peak concurrency of 6 participants, each with up to 2 accompanying vehicles, plus 4 for screening visits
Staff	12	8.5 by 18 feet	11 roles, of which at most 9 are on site at once, plus 3 for monitors and vendors
Accessible	4	9 by 18 feet, one van-accessible at 11 feet	Above the code minimum for 46 stalls; an oncology population justifies the surplus
Electric vehicle	3	8.5 by 18 feet	Title 24 new construction and alteration provisions [51]
Drop-off, short term	3	8.5 by 20 feet	20-minute limit; adjacent to the canopy
Loading and service	2	10 by 25 feet	Deliveries and waste pickup under document 09 §5
Total	**46**		Front matter site table

Table 30. *The stall schedule. The count column sums to the 46 stalls fixed in the front matter site table, and the basis column shows the arithmetic rather than citing a ratio.*

3. Accessible Parking and Path of Travel

(a) Four accessible stalls are provided, one of which is van-accessible, which exceeds the minimum required for a 46-stall facility under the 2010 ADA Standards for Accessible Design and Title 24 [56, 51].

(b) The accessible route runs from the accessible stalls to the canopy, into the lobby, and to the clinical zone entrance, with a running slope not exceeding 1:20 and a cross slope not exceeding 1:48, a minimum clear width of 48 inches, and no step or lip exceeding one-half inch.

(c) The route is signed at each decision point, and the signage is the wayfinding signage required by document 09 §6. There is one route, described once, in two documents.

(d) Every door on the accessible route has a maximum opening force of 5 pounds and a clear width of not less than 32 inches.

4. Drop-Off, Ambulance, and Loading Geometry

Element	Requirement	Reason
Canopy vertical clearance	14 feet 6 inches	A Type III ambulance with a roof-mounted air conditioning unit clears at 12 feet 6 inches; the margin covers a raised gurney
Ambulance approach	Separate from the participant drop-off lane	A transfer must not be blocked by a waiting vehicle
Turning radius, inside	28 feet	Ambulance and the largest delivery vehicle
Drop-off lane length	60 feet, two vehicles	Peak concurrency of 6 participants staggered at 30-minute intervals
Loading dock	At grade, with a levelling plate	Robot crate delivery during move-in; regulated waste pickup thereafter
Exterior queuing	Prohibited	Document 05 §5; all waiting is interior

Table 31. *Six geometry requirements. The ambulance approach is separated from the drop-off lane for one reason: the evacuation route in document 11 §2 uses the same approach, and a route that can be blocked by a parked car is not a route.*

5. Post-Procedure Transportation

A participant discharged after a pancreaticoduodenectomy does not drive. The requirement is absolute, and the site does not treat it as advice.

(a) Before a procedure is scheduled, the research nurse and participant navigator confirms and records the participant's transportation arrangement for the day of discharge, naming the driver or the service.

(b) The confirmation is recorded in the visit record. A procedure is not scheduled without it.

(c) On the day of discharge, the arrangement is re-confirmed before the discharge order is written.

(d) Where the arrangement fails on the day, the site provides a chartered medical transport at its own cost, funded from the participant cost line in document 15 §1. The alternative, discharging a participant into a rideshare after a major abdominal resection, is not available.

(e) Where a participant has no arrangement available for any scheduled visit, the navigator arranges transport in advance rather than at the door, and the cost is charged to the same line.

The participant-facing framing here follows the patient advocacy work: a right that depends on a participant arranging something difficult, unaided, is a right in name [25].

6. Transportation Demand Management and Monitoring

Measure	Target	Evidence
Staggered appointment scheduling	No more than 6 participant vehicles on site at once	The appointment book, sampled monthly
Staff transit subsidy	Offered to all 11 roles	Payroll record
Secure bicycle parking	6 spaces	As-built drawing
Electric vehicle charging	3 stalls, shared	Utilization log
Annual monitoring report	Filed with the City	The report, on the interval set by document 05 §6

Table 32. *Five demand management measures and the evidence for each. The reporting interval is the interval in document 05 §6, so the City receives one report rather than two on different schedules.*

DOCUMENT 11 OF 15

PDAC LLM Oncology Clinical Trial Site Emergency Preparedness and Business Continuity Plan

1. Scope and Command Structure

This plan covers fire, seismic events, power and utility loss, cybersecurity incidents, model faults, and robot faults, and the continuity of operations after any of them. It uses the eleven roles named in document 14 §1 and creates no new titles.

Function	Held by	Succession
Incident commander	Director of clinical operations	Regulatory affairs and quality manager, then chief executive
Clinical command	Site principal investigator	Sub-investigator, medical oncology
Robot and facility safety	Site safety officer	Robotics vendor field engineer, on call
Model and data	Model governance lead	Data manager and biostatistician
Participant liaison	Research nurse and participant navigator	Lead clinical research coordinator

Table 33. *Five command functions and their succession. Only one decision may not be delegated: the decision to stop a procedure already in progress belongs to the site principal investigator or, in that person's absence, to the sub-investigator, and to nobody else.*

2. Fire and Evacuation

(a) On alarm, all robot instances receive a hardware stop under document 09 §3 and are left in place. No attempt is made to power down a robot in an orderly sequence during an alarm.

(b) Egress follows the routes established in the life safety plan approved at commissioning gate C5 [53]. There are two means of egress from each procedure suite.

(c) A participant under anesthesia is moved by defend-in-place to the adjacent smoke compartment, and is evacuated from the building only on the order of the incident commander after consultation with clinical command.

(d) Assembly is at the designated point outside the loading area. The ambulance approach described in document 10 §4 is kept clear at all times and is not used for assembly.

(e) Accountability is by badge reader roll call for the restricted and machine zones, and by sign-in sheet for the clinical zone.

3. Seismic and Structural Events

(a) During shaking, personnel take cover and no procedure is started. A procedure already in progress is stopped at the earliest safe point on the order of clinical command.

(b) After shaking stops, the site safety officer performs a documented walk of every restricted zone before any robot instance is re-energized.

(c) Robotic operations may not resume until every instance in the affected suite has passed an interlock test and an envelope survey, and the robot instance register has been amended under document 06 §4. A seismic event is a register-amending event even where no fault is found, because the absence of a fault is itself the finding that has to be recorded.

(d) Structural inspection by a licensed engineer is required before re-occupancy where the recorded ground motion exceeds the threshold in the facility's post-event inspection procedure.

4. Power Failure and Utility Loss

System	Backup	Runtime	Action at exhaustion
Procedure suite, critical branch	Uninterruptible supply, then generator	15 minutes bridge, 96 hours generator	Close the case; transfer under the mutual aid arrangement in §6
Machine room	Uninterruptible supply, then generator	30 minutes bridge, 96 hours generator	Orderly model shutdown; the human-only pathway continues
Specimen freezers	Generator, plus carbon dioxide backup	96 hours, plus 24 hours	Transfer specimens to the partner site under the mutual aid arrangement
Pharmacy containment exhaust	Generator	96 hours	Suspend compounding; no exception
Life safety branch	Generator	96 hours	Evacuate under §2
Data and audit trail	Uninterruptible supply, then generator	Recovery point 15 minutes, recovery time 4 hours	Restore from the geographically separate replica

Table 34. *Six systems, their backup and their runtime. The 96-hour generator figure assumes the standing fuel contract; without it the figure is the tank, which is 30 hours, and the plan states both rather than the more favorable one.*

5. Cybersecurity Incident and Model Fault

These are two distinct events with two distinct responses, and treating them as one is the error this section exists to prevent.

Event	First action	What continues
Network compromise, clinical segment	Isolate the segment; notify the incident commander within 15 minutes	Procedures continue on paper source documents; nothing is entered until integrity is verified
Network compromise, model segment	Isolate; suspend all model output	Everything. The site reverts to the human-only pathway, which is the standard of care
Model output outside the verified envelope	Suspend the model version; record the fault	Everything, on the human-only pathway
Audit trail chain integrity failure	Freeze the store; preserve; notify the department within 24 hours	Procedures continue; no completion is accepted until the chain is restored
Robot fault or unintended motion	Hardware stop; secure the suite; record a reportable robotic event	The suite is closed until an interlock test passes

Table 35. *Five incident classes. The second and third rows carry the whole design: a model fault does not stop the trial, it stops the advisory output, because the human pathway was never dependent on the model.*

Response follows the identify, protect, detect, respond, and recover functions of the NIST Cybersecurity Framework 2.0 [57]. The fault taxonomy for model output is the verification-before-generation taxonomy established in the 2026 legislative work [36]. A model fault is reported to the department under document 06 §6 and appears in the department's annual report to the Legislature under document 01 §6.

6. Business Continuity, Drills, and Training

(a) The recovery point objective for trial data is 15 minutes and the recovery time objective is 4 hours, achieved by continuous replication to a geographically separate store.

(b) The site maintains a written mutual aid arrangement with at least one clinical facility within 30 minutes travel time for participant transfer, and with one specimen storage facility. The intended partner of choice sits 1.6 miles away, but no agreement exists, so the arrangement is with whichever facility executes one.

(c) Drills are conducted quarterly on a fixed rotation: quarter one, robot fault during an active procedure; quarter two, cybersecurity incident and model fault; quarter three, building evacuation with a participant under anesthesia; quarter four, extended power failure and continuity activation.

(d) Every drill produces an after-action record reviewed by the incident commander, and any corrective action enters the quality system under document 12 §6.

(e) Emergency preparedness training is 8 hours annually for all personnel, and is part of the training hours fixed in document 12 §5. This document does not set its own hours.

DOCUMENT 12 OF 15

PDAC LLM Oncology Clinical Trial Site Activation Checklist and Standard Operating Procedures

1. Activation Gates

Five gates. Each carries a stop condition that is a signed instrument or a number rather than a judgment, which is the gate discipline the capitalization plan established for its twelve milestones [58].

Gate	Name	Stop condition, stated before the work begins
G1	Premises accepted	Commissioning gate C8 closed and the certificate of occupancy issued under document 08 §6
G2	Systems verified	All 14 robot instances passed an interlock test; the segmentation test at C6 failed to make a connection; the audit trail chain verified
G3	People in place	All 11 roles filled; delegation log signed; training hours in §5 complete for every named person
G4	Regulatory cleared	Institutional review board approval in hand; safe-to-proceed on the investigational new drug application; state authorization under document 01 §3
G5	First participant in	G1 through G4 all closed; site initiation visit complete; first participant consented on the approved form

Table 36. *The five activation gates. G1 opens where commissioning gate C8 closes, so the building document and this one share a boundary rather than overlapping, and no work is counted twice.*

2. The Activation Checklist

☐	Item	Owner	Evidence that closes it
☐	Certificate of occupancy issued	Director of clinical operations	The certificate
☐	Air balance report accepted	Site safety officer	Balance report against document 08 §3
☐	Medical gas verification	Site safety officer	NFPA 99 verification certificate
☐	Generator load bank and transfer test	Site safety officer	Test record, 96-hour fuel contract executed
☐	Network segmentation test	Model governance lead	Failed connection attempt log
☐	Envelope survey, 14 instances	Site safety officer	Survey record and floor markings photographed
☐	Interlock test, 14 instances	Site safety officer	Test record filed to the robot instance register

☐	Item	Owner	Evidence that closes it
☐	Robot instance register filed	Regulatory affairs and quality manager	Department filing receipt
☐	Model version register entry	Model governance lead	Register row with content hash
☐	Audit trail chain verified	Model governance lead	Verification record
☐	Pharmacy compounding competency	Investigational drug pharmacist	USP 797 and 800 competency records
☐	Laboratory certification confirmed	Data manager and biostatistician	Clinical Laboratory Improvement Amendments certificate [59]
☐	Delegation of authority log signed	Site principal investigator	The signed log
☐	Training hours complete, all 11 roles	Director of clinical operations	Training matrix at 100 percent
☐	Standard operating procedures approved	Regulatory affairs and quality manager	The index at §3, every row approved
☐	Institutional review board approval	Regulatory affairs and quality manager	The approval letter
☐	Safe-to-proceed confirmed	Chief executive and sponsor representative	The agency correspondence
☐	State authorization issued	Regulatory affairs and quality manager	The authorization
☐	Transportation arrangements procedure live	Research nurse and participant navigator	Document 10 §5 procedure, approved
☐	Site initiation visit complete	Director of clinical operations	The visit report

Table 37. *Twenty checklist items. Every owner is one of the eleven roles in document 14 §1 and never a department name, because a department cannot be asked why an item is still open.*

3. Standard Operating Procedure Index

Class	Procedure	Owner	Review
Conventional	Informed consent and re-consent	Lead clinical research coordinator	Annual
Conventional	Eligibility screening and enrollment	Lead clinical research coordinator	Annual
Conventional	Dose escalation decision and cohort review	Data manager and biostatistician	Annual
Conventional	Investigational product receipt, storage, dispensing, return	Investigational drug pharmacist	Annual
Conventional	Specimen collection, processing, chain of custody	Data manager and biostatistician	Annual
Conventional	Adverse event capture, grading, reporting	Regulatory affairs and quality manager	Annual
Conventional	Source documentation and correction	Lead clinical research coordinator	Annual
Conventional	Monitoring visit hosting and query resolution	Director of clinical operations	Annual
Conventional	Protocol deviation identification and reporting	Regulatory affairs and quality manager	Annual
Conventional	Archive and retention	Regulatory affairs and quality manager	Biennial
Model	Model version registration and release	Model governance lead	Semiannual
Model	Prompt and completion disposition recording	Model governance lead	Semiannual
Model	Audit trail chain verification	Model governance lead	Semiannual

Class	Procedure	Owner	Review
Model	Human-only pathway on participant request	Research nurse and participant navigator	Semiannual
Model	Model fault identification and suspension	Model governance lead	Semiannual
Robot	Envelope survey and floor marking	Site safety officer	Semiannual
Robot	Interlock test and register amendment	Site safety officer	Semiannual
Robot	Emergency stop actuation and two-person restart	Site safety officer	Semiannual
Robot	Terminal cleaning of a procedure suite	Site safety officer	Annual
Robot	Reportable robotic event capture and notification	Regulatory affairs and quality manager	Semiannual

Table 38. *Twenty procedures in three classes. The class column exists so that a monitor can see at a glance which procedures a conventional Phase 1 site would also hold, which is the first ten, and which exist only because of the model and the robots.*

4. Delegation of Authority

(a) The delegation of authority log records, for each person: name, role, delegated tasks by identifier, training completion date, signature of the site principal investigator, effective date, and end date.

(b) A task may not be performed before the log entry authorizing it exists. This rule has no exception, including during activation.

(c) The delegated task sets are listed by role in document 14 §2, and the log refers to them by identifier rather than restating them.

(d) The log is reviewed at every monitoring visit and at every departmental survey.

5. Training and Competency

This section owns the training hours. Documents 06 §3 and 11 §6 cite them and do not restate them.

Role	Initial hours	Competency assessment
Site safety officer	160	Interlock test performed under observation; emergency stop timing measured
Model governance lead	120	Register entry, chain verification, and a fault suspension performed under observation
Lead clinical research coordinator	80	Consent discussion observed and scored
Director of clinical operations	80	Activation checklist walkthrough
Regulatory affairs and quality manager	80	Safety report drafted and reviewed
Research nurse and participant navigator	80	Consent observation; transportation procedure walkthrough
Investigational drug pharmacist	60	Compounding competency under USP 797 and 800
Data manager and biostatistician	60	Escalation decision dry run
Site principal investigator and sub-investigator	40	Protocol examination
Chief executive and sponsor representative	40	Sponsor obligations examination
All personnel, annually	24	Refresher, including 8 hours emergency preparedness

Table 39. *Initial training hours by role and the competency assessment that closes each. Re-training is triggered by a role change, a Class A or Class B deficiency in that role's area, or a failed competency assessment.*

6. Quality Management and Deviation Handling

(a) Deviations are classified as Class A, Class B, or Class C with the meanings established in document 01 §5. A single event is graded once, on one scale, and the classification is recorded with the reason.

(b) A Class A deviation is reported to the department within 24 hours under document 06 §6 and enters the corrective and preventive action system immediately. A Class B deviation is reported within 5 business days. A Class C deviation is reported in the annual report.

(c) Every corrective action names an owner, a due date, and the evidence that will close it. An action with no evidence field is not accepted into the system.

(d) A quality review is held quarterly, attended by the director of clinical operations, the regulatory affairs and quality manager, the site safety officer, and the model governance lead. Its record is the input to the annual report under document 06 §6.

DOCUMENT 13 OF 15

Conventional Pancreatic Cancer Clinical Trial Requirements Manual

1. Scope: What the Site Must Have Regardless of the Model

This document is written to pass one test. A reader who deletes every large language model and every robot from the site still holds a complete conventional Phase 1 pancreatic cancer requirements manual. Nothing in the six sections that follow depends on the model, and nothing in them is relaxed because the model is present.

That order matters. A site that satisfies its model governance obligations and falls short on investigational product accountability has failed as a clinical trial site, and no amount of audit trail repairs it. The model earns its place inside the conventional requirements; the conventional requirements do not bend to accommodate it.

2. Protocol, IND, and IRB

The trial is an open-label, single-arm, 3+3 dose escalation of perioperative daraxonrasib with a staged robotic pancreaticoduodenectomy in resectable pancreatic ductal adenocarcinoma, enrolling up to 18 treated participants and up to 30 screened [3]. The filing is the investigational new drug application deposited in July 2026 [22], written against the section skeleton of the adapted Part 312 framework [16, 23].

Three positioning statements travel with every description of this trial and are stated here rather than left to a reader to infer. Daraxonrasib is investigational and is already in Phase 3 evaluation; it is not described as first in human anywhere in this package, and the supportable novelty claim concerns the integrated surgical and advisory workflow. The robotic configuration is specified at the site agreement and not before. No drug supply agreement, letter of authorization, or regulatory cross-reference is in place with the agent's developer, and no approach has been made.

Institutional review board review is obtained before any screening activity. The board of record is the board named in the site agreement. Continuing review, amendment submission, and reporting of unanticipated problems follow the adapted human subject protection framework [15, 28, 29].

3. Eligibility, Consent, and Enrollment

Criterion	Statement
Inclusion	Histologically or cytologically confirmed pancreatic ductal adenocarcinoma
Inclusion	Resectable disease by multidisciplinary review, without arterial involvement
Inclusion	Age 18 years or older
Inclusion	Eastern Cooperative Oncology Group performance status 0 or 1
Inclusion	Adequate hematologic, hepatic, renal, and coagulation function by protocol thresholds
Inclusion	Able to give written informed consent and to attend the visit schedule
Exclusion	Prior systemic therapy for pancreatic ductal adenocarcinoma
Exclusion	Borderline resectable or locally advanced disease
Exclusion	Distant metastatic disease on baseline imaging
Exclusion	Uncontrolled intercurrent illness, including active infection requiring systemic therapy
Exclusion	Clinically significant cardiac disease, or a corrected QT interval above the protocol threshold
Exclusion	Pregnancy or unwillingness to use protocol-specified contraception

Table 40. *Six inclusion and six exclusion criteria. The second exclusion is the one that distinguishes this trial from the external readout it is compared against: RASolute 302 studied metastatic, previously treated disease and is silent on the resectable setting [9].*

Consent follows the architecture in document 02 §4 and is not restated. The enrollment log records screening number, consent date, eligibility disposition, assigned dose level, and, where a participant screens out, the criterion that was not met.

4. Dose Escalation and Dose-Limiting Toxicity

Level	Daily dose	Participants	Escalation rule
−1	100 mg once daily	3 to 6	Entered only on de-escalation from level 1
1	160 mg once daily	3 to 6	0 of 3 with a dose-limiting toxicity escalates; 1 of 3 expands to 6
2	220 mg once daily	3 to 6	As level 1; 2 or more of 6 stops escalation
3	300 mg once daily	3 to 6	As level 1; the highest level studied

Table 41. *Four dose levels and the escalation rule at each. Three escalation levels at up to six participants each is the eighteen-participant ceiling; level −1 is entered only by de-escalation and does not add to it.*

The dose-limiting toxicity window runs from the first dose of daraxonrasib through postoperative day 30. A design that leaves the window unstated is not a design, and a perioperative trial whose window ends before the operation is not measuring the risk it exists to measure.

Grading instrument	What it grades
CTCAE version 5.0 [60]	Every non-surgical adverse event; Grade 3 or higher within the window is presumptively dose limiting
Clavien-Dindo [61]	Surgical complications; Grade IIIb or higher within the window is presumptively dose limiting
ISGPS 2016 definition [62]	Postoperative pancreatic fistula; Grade B or C is recorded and adjudicated
RECIST 1.1 [55]	Radiographic response, where a target lesion is measurable

Table 42. *Four grading instruments and what each governs. Naming all four is what allows a surgical trial to be escalated on a rule rather than on a discussion, because a complication has a grade before it has an opinion.*

Cohort review is convened by the data manager and biostatistician within 5 business days of the last participant in a cohort completing the window. The review record states each participant's dose-limiting toxicity status, the grading instrument applied, and the escalation decision.

5. Pharmacy, Laboratory, Imaging, and Biospecimens

Pharmacy. Investigational product is received, stored under monitored conditions, dispensed against a written order, reconciled at every visit, and returned or destroyed with a record. Accountability follows the sponsor and investigator recordkeeping requirements [23]. Compounding is performed in the buffer room and the containment area built to the air change and pressure requirements in document 08 §3, under USP General Chapters 797 and 800 [50].

Laboratory. Safety laboratory testing is performed by a laboratory certified under the Clinical Laboratory Improvement Amendments [59]. The site's own specimen processing laboratory performs collection, processing, aliquoting, and storage only, and reports no clinical result.

Imaging. Baseline cross-sectional imaging is obtained within 28 days of the first dose. Response assessment follows RECIST 1.1 [55]. Post-treatment imaging follows the protocol schedule. Imaging is performed off site under the imaging agreement, which is why document 08 §4 provides no imaging suite and no shielding.

Biospecimens. Tumor tissue, plasma for circulating tumor deoxyribonucleic acid, and whole blood are collected at protocol-specified time points. Chain of custody is recorded from collection through processing, storage, and shipment. Specimen freezers are on the generator and carbon dioxide backup described in document 11 §4.

6. Monitoring, Source Documentation, Safety Reporting, and Archive

Obligation	Interval or clock	Authority
Site initiation visit	Before first participant	Activation gate G5, document 12 §1
Monitoring visit	Every 6 weeks while enrolling; every 12 weeks otherwise	Risk-proportionate monitoring [14]
Source data verification	100 percent for consent, eligibility, dose, and dose-limiting toxicity; risk-based otherwise	Adapted good clinical practice [14]
Serious and unexpected suspected adverse reaction	15 calendar days	21 CFR 312.32 [23]
Fatal or life-threatening	7 calendar days	21 CFR 312.32 [23]
Reportable robotic event	5 business days; 24 hours with injury	Document 02 §6
Annual report to the agency	Within 60 days of the anniversary	21 CFR 312.33 [23]
Archive	6 years from database lock	Document 03 §5, the longest applicable period

Table 43. *Eight obligations with the clock and the authority for each. The state and federal clocks are compared in document 02 §6 and the shorter governs; nothing in this table is the longer of two.*

Source documentation is attributable, legible, contemporaneous, original, and accurate. A correction strikes through, initials, dates, and states a reason, and never obscures the original entry. Electronic source records, including the prompts and completions treated as source under document 03 §5, meet the electronic record requirements [32].

7. The Evidence Behind the Design

Source	Date	Test arm	Control	Stated limitation, in the authors' own words
Empirical triplicate, 100,000 patients [63]	Jul 2025	Grade 3+ adverse events 8.0%	Grade 3+ 25.0%	One log file favors experimental while the KRAS-mutant report favors control; trial-to-trial variability
QSP simulation, 10 arms [10]	Aug 2025	12.8 months median overall survival, hazard ratio 0.25	5.4 months	Assumes no acquired resistance and ideal pharmacodynamics; Grade 3+ adverse events consistently high in all arms
Digital twin, 1000 patients [11]	Oct 2025	12.1 months median overall survival, progression-free hazard ratio 0.31	Not applicable	No patient-specific pharmacokinetic, pharmacodynamic, or tumor growth parameters; simple maximum-effect model, no immune compartments
RASolute 302 [9]	May 2026	13.2 months median overall survival	6.6 months	Metastatic and previously treated; silent on the resectable setting

Table 44. *Four sources with the limitation each set of authors stated, carried in the same row as the result. No row can be read without the caveat that qualifies it, which is the condition on citing any of them.*

The 2025 simulated ratio of 2.4-fold and the 2026 observed ratio of 2.0-fold are close. That proximity is a chronology observation and a hypothesis-supporting one. It is **not** a validation claim, and three differences are material: 1000 simulated against 241 enrolled; a combination against a single agent; and KRAS G12C against a primarily G12D and G12V population. Those three clauses appear here in the same paragraph as the numbers, and they appear in the same paragraph as the numbers everywhere else in this package.

A credibility assessment aligned to ASME V&V 40 and to ICH M15 returned a score of 81.9 across 55 verification notebook tests [11, 12, 13]. That is a measured figure from the author's own verification suite and is not a regulatory finding.

Work product	Industry benchmark	Industry time	This author
Empirical virtual trial, triplicate	Above $120,000 per run	4.5 months	$36,330 projected, about one month
Ten-arm QSP virtual trial	Above $2,000,000	Several months to over a year	$36,330 projected, about one month
Digital-twin simulator platform	$28,000 to $700,000	3 to 16 weeks	$36,330 projected, about one month

Table 45. *Three work products against their industry benchmarks. The $36,330 figure is a projection from the author's own source set, computed at one operator working sixty hours a week, and is reported as projected rather than as a completed accounting.*

The reason a funder should read this section rather than the model governance sections is that the design decisions above are the ones a data monitoring committee will examine: a stated dose-limiting toxicity window that spans the operation, four named grading instruments, an eighteen-participant ceiling that follows from the escalation rule rather than from a budget, and four evidence sources each carrying its own stated limitation.

DOCUMENT 14 OF 15

Staffing, Role Delineation, and Move-In Plan for a Chief Executive and Ten Coworkers

1. The Eleven Roles

No	Role	FTE	The one thing the site cannot do without this role
1	Chief Executive Officer and sponsor representative	0.20	Hold the investigational new drug application and sign every regulatory submission
2	Site principal investigator, hepatopancreatobiliary surgery	0.10	Perform the resection and hold participant safety
3	Sub-investigator, gastrointestinal medical oncology	0.10	Set and manage perioperative dosing and toxicity
4	Director of clinical operations	0.40	Close the activation checklist and own the procedure index
5	Lead clinical research coordinator	1.00	Run consent, visits, and source documents every working day
6	Regulatory affairs and quality manager	0.45	Correspond with the board and the agency, and hold the archive
7	Investigational drug pharmacist	0.20	Account for every unit of investigational product
8	Robotics and physical AI systems engineer, site safety officer	0.55	Own the envelopes, the interlocks, and the emergency stop chain
9	LLM verification and model governance lead	0.40	Hold the advisory-only boundary and the audit trail
10	Data manager and biostatistician	0.30	Convene cohort review and lock the database
11	Research nurse and participant navigator	0.25	Carry the participant pathway, including transportation
	Total	**3.95**	

Table 46. *The eleven roles. The column sums to 3.95 full-time equivalents on the award, which is the number document 06 §3 surveys against and the number document 15 §1 prices.*

Eleven roles at 3.95 award-funded full-time equivalents is not eleven salaries. Nine of the eleven are fractional, and the fractions are what make an eleven-person site affordable inside a $700,000 annual direct cost. The balance of each fractional role's time is either contributed by the host institution under the site agreement or is not charged to the award at all.

2. Qualifications and Delegated Tasks

Role	Qualification verified by the department	Delegated task set
Chief executive and sponsor representative	None; not a clinical position	Sponsor obligations, regulatory signature, financial certification. No clinical task
Site principal investigator	California medical license; hepatopancreatobiliary practice; 40 hours	Eligibility confirmation, consent, procedure, dose-limiting toxicity adjudication, delegation log signature
Sub-investigator, medical oncology	California medical license; gastrointestinal oncology; 40 hours	Dosing, toxicity management, consent, adverse event grading
Director of clinical operations	Five years operations; 80 hours	Activation checklist, procedure index, monitoring visit hosting, drill command
Lead clinical research coordinator	Coordinator certification; 80 hours; consent competency	Consent, screening, visit conduct, source documentation, query resolution
Regulatory affairs and quality manager	Three years submissions; 80 hours	Board correspondence, safety reporting, deviation handling, archive
Investigational drug pharmacist	California pharmacist license; USP 797 and 800 competency	Receipt, storage, compounding, dispensing, reconciliation, return
Site safety officer	160 hours; platform certification; interlock competency	Envelope survey, interlock test, emergency stop chain, terminal cleaning oversight
Model governance lead	120 hours; verification competency	Version register, disposition recording, chain verification, fault suspension
Data manager and biostatistician	Statistical qualification; lock competency	Cohort review, escalation decision record, database lock, specimen chain oversight
Research nurse and participant navigator	California registered nurse license; oncology; 80 hours	Participant pathway, human-only pathway requests, transportation, education

Table 47. *Qualification and delegated task set by role. The qualification column is word for word the column in document 06 §3, because a department that surveys against a different standard than the site staffs against will cite a deficiency that nobody can correct.*

3. The Sponsor-Investigator Firewall

The chief executive is not a clinical investigator on this trial. He is chief executive of the sponsor and holds 100 percent of it, so every trigger in 21 CFR 54.2 would fire if he held both roles: a proprietary interest in the tested product, an equity interest in the sponsor, and compensation whose value could be affected by the outcome [24]. The investigator role therefore sits wholly with the site.

A Phase 1 3+3 escalation carries no randomization, so the firewall cannot bind on treatment assignment. It binds instead on the three decisions where an interested party could move the result: dose assignment within the escalation rule, dose-limiting toxicity adjudication, and the analysis plan. Each of those three is held by a role at the site, and none is held by the chief executive. The wording is carried forward from the company's own capitalization work so that it still binds when the Phase 2 successor introduces randomization [58, 64].

Decision	Held by	Why it is firewalled
Dose assignment within the escalation rule	Data manager and biostatistician, on the cohort review record	The only discretionary step in a 3+3 design
Dose-limiting toxicity adjudication	Site principal investigator, with the sub-investigator	Determines whether escalation continues
Analysis plan and database lock	Data manager and biostatistician	Determines what the trial reports
Regulatory signature	Chief executive	Not firewalled; a sponsor obligation that cannot be delegated to the site

Table 48. *Four decisions and where each sits. The fourth row is included because a firewall described only by what it excludes leaves a reader unable to see what the sponsor still does.*

4. Coverage, Call, and Cross-Training

A roster of 3.95 award-funded full-time equivalents cannot cover a twenty-four-hour service, and the site does not attempt one. Coverage is the window in document 05 §5: 06:00 to 20:00 on weekdays and 08:00 to 14:00 on Saturday.

(a) Outside the window, a participant in the postoperative period is covered by the surgical service under the site agreement, and by the site's own on-call clinical line staffed by the site principal investigator or the sub-investigator.

(b) The research nurse and participant navigator carries the participant- facing on-call line during the window and hands it to the clinical line outside it.

(c) Cross-training pairs are fixed so that no single absence closes the site: the lead clinical research coordinator and the research nurse cross-train on consent and visit conduct; the director of clinical operations and the regulatory affairs and quality manager cross-train on deviation handling; the site safety officer and the model governance lead cross-train on the segmentation test and the audit trail chain verification.

(d) A cross-trained person performs a task only where the delegation log authorizes it, under document 12 §4. Cross-training is a prerequisite for delegation and never a substitute for it.

5. Cost of the Roster

Role	FTE	Per year	Note
Chief executive and sponsor representative	0.20	$42,000	Sponsor obligations only
Site principal investigator	0.10	$34,000	Surgical rate; the balance is clinical time not charged
Sub-investigator, medical oncology	0.10	$28,000	As above
Director of clinical operations	0.40	$54,000	
Lead clinical research coordinator	1.00	$88,000	The only full-time award-funded role
Regulatory affairs and quality manager	0.45	$52,000	
Investigational drug pharmacist	0.20	$30,000	
Site safety officer	0.55	$72,000	Includes the platform certification maintenance
Model governance lead	0.40	$58,000	
Data manager and biostatistician	0.30	$38,000	
Research nurse and participant navigator	0.25	$25,000	
Total	**3.95**	**$521,000**	Salary and fringe combined

Table 49. *The cost of the roster. The charged column sums to $521,000, and the remaining $179,000 of the $700,000 annual direct cost is accounted for line by line in document 15 §1.*

The $700,000 per year and $3,500,000 over five years frame is reused verbatim from the ten-application work and is not re-derived here [8, 39]. The arithmetic that splits it is simple and is shown rather than asserted: $521,000 of personnel plus $179,000 of non-personnel is $700,000, and five years of that is $3,500,000.

6. The Move-In Schedule

The schedule below closes at week 46 only because four of the nine permits in document 05 §6 run in parallel. Read strictly sequentially, the same nine permits add roughly eleven weeks.

Weeks	Activity	Role entering, and why then
1 to 2	Lease execution, zoning verification	1, chief executive: signs the lease and the application
3 to 6	Schematic design; permit application filed	4, director of clinical operations: owns the schedule from here
7 to 14	Hearing; plan check submittal	
15 to 20	Permits issued; demolition	6, regulatory affairs and quality manager: begins the board and agency file
21 to 32	Construction	8, site safety officer: witnesses rough-in and anchorage
33 to 36	Robot delivery, integration, envelope survey	2 and 3, investigators: review suite layout against the procedure
37 to 38	Commissioning gates C1 through C7	9, model governance lead: runs the segmentation test at C6
39	Certificate of occupancy; gate G1	7, pharmacist: takes possession of the pharmacy
40 to 42	Onboarding, training, procedure approval; gate G3	5, lead coordinator; 11, research nurse and navigator
43	Board approval and safe-to-proceed confirmed; gate G4	10, data manager and biostatistician
44	Site initiation visit	
45	Activation review; gates G1 to G4 confirmed closed	
46	First participant consented; gate G5	All 11 roles on site

Table 50. *The forty-six-week move-in. Roles enter on the week each is first needed rather than at a single start date, which is what keeps the personnel line inside $521,000 in year one.*

DOCUMENT 15 OF 15

Federal Funding Stewardship, Lobbying, and Legislative Engagement Plan

1. The Award and Its Terms

The funding position assumed throughout this package is a federal award of $700,000 per year for five years, $3,500,000 in direct cost, from at least one federal agency, with no cost share and no program income anticipated. The frame is reused from the ten-application work and is not re-derived [8, 39].

Line	Per year	Five years	Basis
Personnel, 3.95 award-funded full-time equivalents	$521,000	$2,605,000	Document 14 §5
Premises and occupancy, La Jolla clinical suite	$96,000	$480,000	12,400 square feet at the leasehold rate, plus common area
Physical AI and on-premises compute	$38,000	$190,000	Machine room, model node, interlock rig, logging
Regulatory, institutional review board, and archive	$21,000	$105,000	Board fees, filings, six-year archive
Participant costs not covered by standard of care	$14,000	$70,000	Transportation under document 10 §5, parking validation, stipends
Verification, monitoring, and audit	$10,000	$50,000	Monitoring visits, envelope surveys, chain verification
Total direct cost	**$700,000**	**$3,500,000**	Sum of the six lines above

Table 51. *The award, line by line. The six lines sum to $700,000 exactly and the personnel line reconciles to document 14 §5 exactly; both sums are checked rather than asserted.*

Indirect cost is not claimed against this award. The company's own analysis of the same direct work routed through a university at a 57 percent facilities and administrative rate showed a 33.1 percent premium to the funder for identical work, and the position taken here follows from that finding rather than from generosity [58, 65]. Where the company later takes a small-business award, the ownership and control test applies and the federal share sits below that firewall [66]. Drug supply, operating theater time, pathology, and bioanalytical support sit in a contributed non-federal column that carries no dollar figure, because no agreement exists and an invented cost-share number is worse than none.

2. Stewardship Controls

Control	Frequency	Record that evidences it
Separation of duties, requisition from approval	Every transaction	The approval trail; no person approves a payment they requested
Prior approval threshold	Every item above $5,000	Written approval by the chief executive, filed
Effort certification	Semiannual	Certified effort report for all 3.95 full-time equivalents
Subaward and vendor monitoring	Annual	Vendor risk assessment and invoice reconciliation
Equipment tagging and inventory	Annual	Tag register, reconciled to the robot instance register in document 06 §4
Financial report	Annual	The federal financial report and the reconciliation to the six lines above
Financial disclosure certification	At submission and annually	Certification and disclosure under 21 CFR 54.4 [24]

Table 52. *Seven stewardship controls. The fifth row ties the financial inventory to the regulatory register, so a robot cannot be on one list and absent from the other.*

3. Lobbying: What Is Permitted and What Is Not

This section has to be exact, because the same activity is lawful with one source of funds and unlawful with another.

Federal award funds may not be used for lobbying. The cost principle is explicit: costs of attempting to influence the outcomes of any federal, state, or local election, and costs associated with certain lobbying activities, are unallowable [67]. Separately, appropriated funds may not be used to pay any person to influence an officer or employee of an agency in connection with the award, extension, or modification of a federal contract, grant, loan, or cooperative agreement, and a recipient must disclose any non-federal lobbying it conducts on those matters [68].

Activity	Funding source	Authority
Technical assistance provided at the written request of a legislature or a committee	Federal award funds permitted	Excepted from the lobbying cost principle [67]
Responding to a documented agency request for factual information	Federal award funds permitted	Excepted; a response to a request is not an attempt to influence
Publishing this package, and depositing it under a digital object identifier	Federal award funds permitted	Dissemination of research results
Advocacy for enactment of documents 01 through 04	Non-federal funds only	Unallowable against the award [67]
Retaining a registered lobbyist	Non-federal funds only	Unallowable against the award; disclosure required [68]
Any attempt to influence a federal award decision	Prohibited from appropriated funds; non-federal funds must be disclosed	31 U.S.C. 1352 [68]
Campaign or election activity	Prohibited from award funds entirely	[67]

Table 53. *Activity, source, and authority. No row shows a federal source beside an advocacy activity. A plan that does not separate the two is not usable, because the separation is what an auditor tests.*

The practical consequence is that three of the four legislative instruments in this package can be published, deposited, and provided on request with award funds, and none of them can be advocated for with award funds. The company maintains a separate non-federal account for the fourth activity, and the accounting separation is a stewardship control under §2.

4. The Legislative Engagement Plan

Instrument	Body and committee function	Artifact shared, and the sequence
Document 01, SB 1188	California Senate, health committee of jurisdiction	The full text plus documents 06 and 12, because the bill is unreadable without the regulation and the checklist it presumes
Document 02, AB 3162	California Assembly, health committee of jurisdiction	The full text plus document 09 §3, which holds the safety numbers the bill restates
Document 03, SB 964	California Senate, privacy and consumer protection committee of jurisdiction	The full text plus document 07 §5, which shows the federal change control the state register serves
Document 04, H. R. 10412	United States House, energy and commerce committee of jurisdiction	The full text plus the predecessor bill H. R. 9510 [35] and the policy basis [5]

Table 54. *Four instruments and the shortest defensible package for each. Committees are named by function rather than by member, because a member changes and a jurisdiction does not.*

The federal engagement rests on a policy record that already exists. The White House report asks the federal research portfolio to prioritize the individual scientist over legacy institutions, and its annexed budget priorities memorandum asks agencies for long-duration grants and non-federal cost share [5, 6]. Two executive orders name robotics and physical artificial intelligence as instruments of discovery and set an evidentiary standard for federally supported science [37, 38]. Document 04 asks for one thing those instruments do not yet provide: a published clock on how long the review of a large language model and robotic workflow takes.

5. Industry and Institutional Communications

What has happened is stated in terms a reader can check, and what has not happened is stated in the same paragraph.

Favorable responses to project applications have been received from federal funders. Three presidential recognition letters have been issued to the chief executive. Industry recipients of the company's communications have been responsive to ChemicalQDevice initiatives. Each of those is a fact about correspondence. None of them is an award, an agreement, a commitment, or a review of any document in this package, and none is presented as one.

Three positioning corrections travel with every communication and are repeated here in full. No drug supply agreement, letter of authorization, or regulatory cross-reference is in place with the agent's developer, and no approach has been made. The robotic configuration is specified at the site agreement and not before. No institutional agreement of any kind exists with UC San Diego, with Moores Cancer Center, or with any other clinical site, and the partner of choice is named at the feasibility stage only.

6. The Author Record That Supports the Ask

ChemicalQDevice has been a California limited liability company since October 2021. Of fifty-six deposited works, thirty-six were generated with artificial intelligence assistance in 2026 alone. Development is conducted in public: demonstrations and sessions run open-source and online alongside other developers, so the method is inspectable rather than asserted. Twenty of the deposited works form the chronology this trial rests on.

Work	Deposited	What the La Jolla site inherits from it
End-to-End PDAC Digital Twin Proposals [69]	Jun 2025	The drug shortlist from which daraxonrasib emerged
100,000 Patient In Silico Triplicate [63]	Jul 2025	The arm screen that preceded enrollment planning
QSP Metastatic PDAC Simulation [10]	Aug 2025	The simulation gate and its stated limitations
Accelerating FDA Compliance via Digital Twin [11]	Sep 2025	The credibility evidence at 81.9 across 55 tests
Adaption: ICH E6(R3) [14]	Mar 2026	The good clinical practice spine in documents 07 and 13
Adaption: 21 CFR Part 50 [15]	Mar 2026	The consent architecture in document 02 §4
Adaption: 21 CFR Part 312 [16]	Mar 2026	The investigational new drug skeleton in document 07
Physical AI Trial Site Documentation [1]	Mar 2026	The eleven-document template this package succeeds
National Platform [33]	Mar 2026	The hash-chained audit trail in document 03 §5
Patient Journey [70]	Mar 2026	The ten-stage participant pathway in document 10 §5
2030: 60 Second Robotic Whipple [54]	May 2026	The device envelope assumed in document 08 §2
Unitree H2 Surgical Humanoid [27]	May 2026	The ten-gate assurance posture in document 09 §3
VVUQ Physical AI Trial Bill [36]	May 2026	Verification before generation, document 07 §5
H. R. 9510 (Bill v5.0) [35]	Jun 2026	The predecessor federal bill document 04 succeeds
Earning the Clinician's Trust [21]	Jun 2026	The supervision rationale in document 07 §5
Phase 1 IND and IDE Protocol [3]	Jun 2026	The protocol the whole site is built to run
Phase 2 Successor Protocol [64]	Jun 2026	The randomization the firewall must survive
Efficient LLM Document Generations [71]	Jun 2026	The document generation method behind this package
Investigational New Drug Application [22]	Jul 2026	The 172-page filing mapped in document 07 §3
Funding Application, first version [72]	Jul 2026	The first statement of the ask, five days before v2.0
Funding Application v2.0 [39]	Jul 2026	The budget frame reused in §1 above
Patient Robot Advocacy [25]	Jul 2026	The participant rights in document 02 §3
10 Funding Applications [8]	Aug 2026	The $700,000 per year frame and the ten mechanisms
Independent Scientist to Novel Performer [58]	Aug 2026	The ownership firewall and the cost benchmarks

Table 55. *Twenty-four deposited works and what this site takes from each. The rows are the argument: the site is not a proposal built on an idea, it is a facility built on a filed protocol, a filed application, and a regulatory spine that already exists in public.*

The company's first years produced no application. A provisional filing did not convert; forty weeks of seminars and a six-month literature review found no scalable use; two years of notebook work concluded that the approach the company had been built around could not escape exponential resource growth as the algorithms scaled. That record is stated because it is why the current method exists. The working method arrived in 2024, when multimodal models could carry text, image, and document context together, and the first full manuscript tested exactly that [7, 73, 74, 75]. Everything in this package is downstream of that change.

LA JOLLA MOVE-IN, BACK MATTER

Back Matter

Abbreviations

Short	Expansion	Short	Expansion
ACH	Air changes per hour	ISGPS	International Study Group of Pancreatic Surgery
ADA	Americans with Disabilities Act	ISO	International Organization for Standardization
AE	Adverse event	LLM	Large language model
CCR	California Code of Regulations	NFPA	National Fire Protection Association
CEQA	California Environmental Quality Act	NIST	National Institute of Standards and Technology
CFR	Code of Federal Regulations	PDAC	Pancreatic ductal adenocarcinoma
CLIA	Clinical Laboratory Improvement Amendments	QSP	Quantitative systems pharmacology
CTCAE	Common Terminology Criteria for Adverse Events	RECIST	Response Evaluation Criteria in Solid Tumors
CUP	Conditional use permit	RPO	Recovery point objective
DLT	Dose-limiting toxicity	RTO	Recovery time objective
DOI	Digital object identifier	SOP	Standard operating procedure
FTE	Full-time equivalent	USP	United States Pharmacopeia
IDE	Investigational device exemption	VVUQ	Verification, validation and uncertainty quantification
IEC	International Electrotechnical Commission	3+3	The dose escalation rule at document 13 §4
IND	Investigational new drug		
IRB	Institutional review board		

Table 56. *Abbreviations, in two column pairs at the body measure. The rule for inclusion is mechanical: every abbreviation used more than once in the body appears here, and every row here is used in the body.*

Data and Code Availability

Every work cited to a digital object identifier is openly deposited. The repository is at https://github.com/kevinkawchak/physical-ai-oncology-trials, version v4.7.0 [76].

The package is deposited in `funding/move-in`. That subtree contains the master prompt filed verbatim and the build output of record in `prompts/`; fifteen stage sub-prompts in three directories under `sub-prompts/`; the three input artifacts in `inputs/` with a README for each; and three complete LaTeX source sets in `draft-move-in/`, `full-move-in/` and `final-move-in/`, each with its own Overleaf bundle. There is no `final-move-in/publication` directory, by instruction.

No raster image is used anywhere in this package. Every rule, badge and mark on the cover is drawn in the style file. The claim is checkable: a search of the subtree for a PNG or a JPEG file returns nothing.

Author Contributions and Conflicts

Kevin Kawchak is the sole author. He is chief executive of ChemicalQDevice and holds 100 percent of it. He is therefore not a clinical investigator on the trial this package describes: every trigger in 21 CFR 54.2 would fire if he held both roles, so the investigator role sits wholly with the site, and the three decisions where an interested party could move the result are held at the site under document 14 §3 [24].

He holds no equity in Revolution Medicines and no equity in any robotic surgery vendor. No sponsor, contract research organization, institutional review board, regulator, or medical society reviewed this package or authorized any statement in it. No agreement of any kind exists with UC San Diego or with Moores Cancer Center. The work was adapted using Claude Code Opus 5.

Citation

Kawchak K. La Jolla Move-In: Pancreatic Oncology Clinical Trial Site Complete Documentation Package. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

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